Mapping Cancer Therapy Mechanisms, Evidence Gaps, and Resistant-State Hypotheses: A Cross-Literature Analysis

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Abstract

Cancer-therapy literatures often mature in parallel without being compared on shared axes such as evidence depth, resistant-state biology, pathway dependencies, or delivery constraints. Here we present a cross-literature analysis of 4,830 full-text cancer research articles (803 journals, 19 mechanisms, 22 cancer types, 2001-2026), alongside a separate archive of 5,584 abstract-only records, to map where research is concentrated, where apparent gaps are likely to be artifacts, and which mechanistic ideas deserve broader comparison. We measure the quality of our own evidence tagger (46% exact-label accuracy, 96% binary precision, 55% recall on a 100-article gold set), derive tissue-of-origin and weighted-evidence layers to sharpen mechanism comparisons, and introduce a diagnostic-to-therapy matching framework covering six translational chains. Monte Carlo and spatial simulations explore ferroptosis biochemistry across cell phenotypes, tissue depths, drug penetration gradients, drug combinations, and oxygen-mediated tumor microenvironment effects. Key simulation findings: (1) RSL3 + FSP1 inhibitor produces 1.99× Bliss synergy through dual-pathway depletion, (2) RSL3 efficacy collapses from 3.7% to 0.1% under tumor hypoxia while SDT maintains 87.8% kill because exogenous ROS bypasses the mitochondrial bottleneck, and (3) tissue-specific drug penetration reduces RSL3 kill from 40% (2D) to 1.8% behind the blood-brain barrier. Within this broader landscape, one specific hypothesis survives as worth testing rather than assuming: physical ROS-generating modalities, especially sonodynamic therapy (SDT), may maintain efficacy in conditions — hypoxia, deep tissue, limited drug penetration — where pharmacologic ferroptosis inducers fail. We frame this as a conditional implication of the wider analysis, not the article's starting assumption. The result is a more cautious but more useful conclusion: the important question is not simply which modality looks interesting, but which interventions remain plausible after accounting for resistance state, tissue access, tumor microenvironment, evidence maturity, and known corpus artifacts. All simulation code (MIT-licensed Rust library with Python bindings), analysis pipelines, and data are openly available for replication and extension.

Keywords: cross-literature analysis, resistant states, evidence gaps, ferroptosis, photodynamic therapy, sonodynamic therapy, radioligand therapy, immunotherapy, pathway targets, cancer therapy

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Part I

Why This Exists

Chapter 1

The Problem

[To be added in #101]

Chapter 2

What We Set Out To Do

[To be added in #101]

Part II

What We Found

Chapter 3

The Literature Landscape

3.1 Introduction

Cancer research is rich in promising mechanisms but poor at comparing them on a common frame. Fields grow around modalities, delivery platforms, or molecular targets, yet the real translational questions often cut across those boundaries: which interventions have credible human evidence, which are trapped in preclinical loops, which resistant states are actually being targeted, and which conclusions are artifacts of search design rather than biology. This problem becomes acute when one literature becomes rhetorically compelling before it has been properly compared against adjacent approaches.

Drug-tolerant persister cells illustrate the problem clearly. These residual populations are a central obstacle to durable remission, and recent work shows that some persister states are sensitized to ferroptosis, an iron-dependent cell-death pathway driven by lethal lipid peroxidation M et al. [2026]. That finding immediately raises a therapeutic question, but not just the obvious one. The field has focused largely on pharmacologic ferroptosis inducers such as RSL3, erastin, and GPX4-directed strategies G et al. [2022]. A broader question is whether resistant-state vulnerabilities should sometimes be matched to different delivery logics entirely, including spatially selective physical modalities, immune-engaging strategies, or pathway-directed approaches that sit outside the original literature silo.

This article therefore takes a broader starting point than our earlier framing. We first map the full-text corpus across 19 mechanism tags and 22 cancer types, assess evidence maturity, and highlight where gap claims are unstable because of taxonomy, missing full text, or incomplete evidence coverage. We then use that broader landscape to evaluate one narrower implication: whether physical ROS-generating modalities such as photodynamic therapy (PDT), sonodynamic therapy (SDT), and more speculatively cold atmospheric plasma (CAP) should be treated as candidate tools for locally targetable, ferroptosis-prone residual states rather than as generic alternatives to drugs. This reframing matters

because PDT/SDT are not the only adjacent directions that emerge when the repo is examined without modality loyalty. Radioligand therapy, phagocytosis checkpoints, targeted protein degradation, stromal shielding, senescence programs, and lipid-remodeling defenses all compete for attention once the analysis is widened.

We present a systematic analysis of 4,830 full-text cancer research articles spanning 19 therapeutic mechanisms and 22 cancer types (803 journals, 2001-2026), with a separate archive of 5,584 abstract-only records, together with simulation work on ferroptosis sensitivity across cell phenotypes. The aim is not to prove that one modality is the answer. It is to identify which hypotheses remain plausible after accounting for evidence depth, known corpus artifacts, and the biology of residual disease.

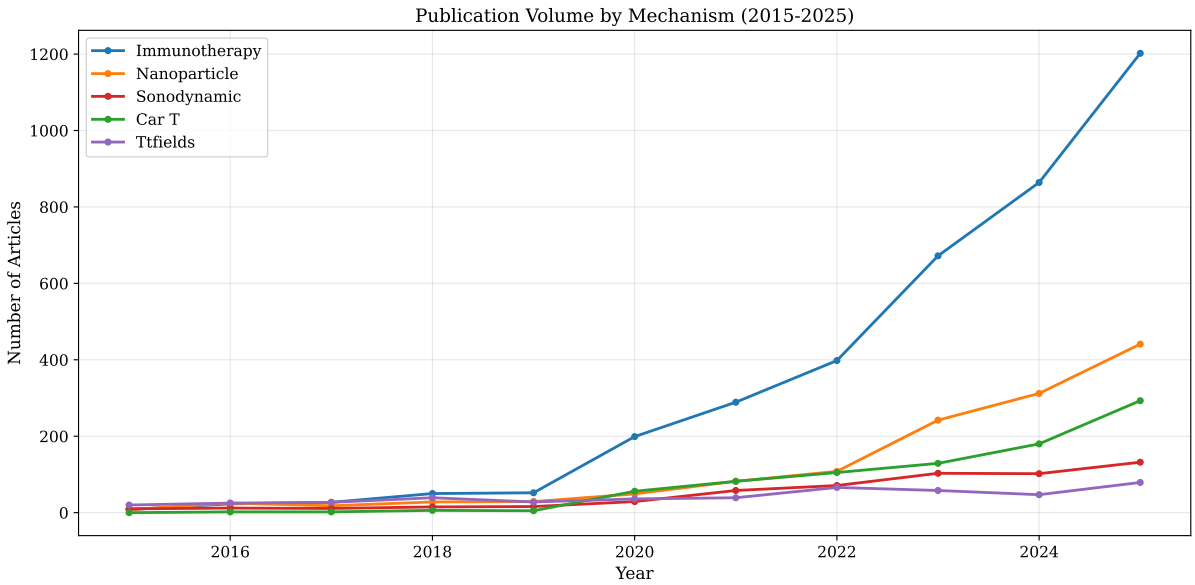

Figure 3.1: Publication volume 2015–2025.

3.2 Search Strategy and Corpus Construction

We conducted a systematic literature search of PubMed using the NCBI E-utilities API, covering publications indexed through March 2026. The search strategy employed mechanism-specific query sets using established MeSH terms and free-text synonyms: for example, "tumor treating fields" OR "TTFields" OR "Optune" for TTFields; "chimeric antigen receptor" OR "CAR-T" OR "CAR T cell" for CAR-T therapy; "sonodynamic therapy" OR "sonosensitizer" for sonodynamic approaches. During revision, we also identified additional underrepresented areas for future corpus expansion, including CAP, radioligand therapy/theranostics, targeted protein degradation, phagocytosis checkpoints, autophagy, senescence/SASP, stromal biology, cuproptosis, and disulfidptosis. The quantitative results reported in this manuscript still refer to the frozen 19-mechanism corpus

analyzed here; the expanded taxonomy is a next-pass scaffold rather than a retroactive redefinition of the present results. Results were enriched with metadata from OpenAlex (open access status, citation counts, topic classification) and PubMed Central or publisher-hosted OA URLs (full-text retrieval for open access articles).

3.3 Inclusion and Exclusion Criteria

Articles were included if they were returned by the PubMed search queries and addressed at least one of the 19 therapeutic mechanisms in the context of cancer treatment or cancer biology. No additional manual inclusion or exclusion filtering was applied beyond the search query constraints. The resulting archive therefore includes original research, clinical trial reports, systematic reviews, narrative reviews, and some case series. For quantitative analyses in this manuscript, we indexed only the 4,830 records for which local full text was available; the remaining 5,584 metadata/abstract-only records were retained in a separate archive and not used for the full-text-derived corpus statistics reported below. This distinction is important: article counts in the Results section refer to the full-text corpus unless explicitly stated otherwise.

3.4 Categorization Framework

Each article was tagged along three axes:

Therapeutic mechanism(s): Articles were assigned one or more of 19 tags representing distinct therapeutic modalities. We use "mechanism" broadly to encompass molecular targets (e.g., synthetic lethality), drug classes (e.g., ADCs), delivery platforms (e.g., nanoparticles), device-based interventions (e.g., TTFields), and biological strategies (e.g., microbiome modulation). This heterogeneous taxonomy reflects the diversity of the contemporary oncology toolkit; we prioritized practical distinctness over strict ontological consistency. The 19 tags are: (1) antibody-drug conjugate (ADC), (2) bioelectric modulation, (3) bispecific antibody, (4) CAR-T cell therapy, (5) CRISPR gene editing, (6) electrochemical therapy (including irreversible electroporation), (7) electrolysis, (8) epigenetic therapy, (9) frequency therapy (including pulsed electromagnetic fields), (10) high-intensity focused ultrasound (HIFU), (11) checkpoint immunotherapy, (12) mRNA cancer vaccine, (13) metabolic targeting, (14) microbiome modulation, (15) nanoparticle delivery, (16) oncolytic virotherapy, (17) sonodynamic therapy, (18) synthetic lethality, and (19) tumor-treating fields (TTFields). Articles addressing multiple mechanisms received multiple tags, enabling convergence analysis.

Cancer type(s): Articles were assigned to one or more of 22 cancer types: bladder, breast, cervical, colorectal, esophageal, gastric, glioblastoma, head-and-neck, kidney,

leukemia, liver, lung, lymphoma, melanoma, mesothelioma, myeloma, neuroblastoma, ovarian, pancreatic, prostate, sarcoma, and thyroid.

Evidence level: Original-research-like articles were classified into one of seven tiers: (1) Phase III clinical trial or randomized controlled trial, (2) Phase II clinical trial, (3) Phase I clinical trial, (4) non-phase clinical / patient-series evidence ('clinical-other'), (5) preclinical in vivo, (6) preclinical in vitro, or (7) theoretical/computational. Review-like articles (reviews, systematic reviews, meta-analyses, evidence maps) and protocols were intentionally left unclassified rather than being forced into an evidence tier. Classification was based on keyword detection plus publication-type metadata for the highest level of evidence reported in the article text. The 'clinical-other' bucket is deliberately conservative: it indicates patient-level evidence exists, but should not be interpreted as equivalent to phase-labeled trial maturity.

3.5 Analytical Approach

Four primary analyses were conducted:

Mechanism-cancer matrix: A 19 x 22 cross-tabulation of article counts by mechanism and cancer type, identifying concentration patterns and gaps.

Convergence map: Identification and quantification of all mechanism-pair co-occurrences across the corpus, revealing which multi-mechanism combinations are most actively researched.

Evidence-tier analysis: Assessment of the highest clinical evidence level achieved for each mechanism, identifying translational bottlenecks.

Gap analysis: Identification of mechanism-cancer and mechanism-mechanism pairs with zero or near-zero co-occurrence in our corpus, noting that some gaps reflect search query design rather than genuine research absence.

Impact metrics (citation counts, iCite Relative Citation Ratio) were used to identify landmark publications and assess field-normalized influence. Publication volume was tracked annually to identify growth trajectories.

3.6 Limitations

Several limitations should be acknowledged. First, all tagging was performed by automated keyword matching against curated dictionaries (no manual article-level review), introducing potential misclassification. A 100-article stratified gold-set evaluation of the evidence tagger measured 46% exact-label accuracy, with binary evidence-detection precision of 96%, recall of 55%, and F1 of 0.701 (Figure 3.2). The tagger therefore behaves as a conservative detector: when it assigns an evidence label, that label is almost always

correct, but it misses roughly half of valid evidence-bearing records — particularly in the clinical-other (23% recall), theoretical (5% recall), and preclinical-in-vitro (35% recall) categories. These metrics should be used to calibrate how absence claims are interpreted throughout.

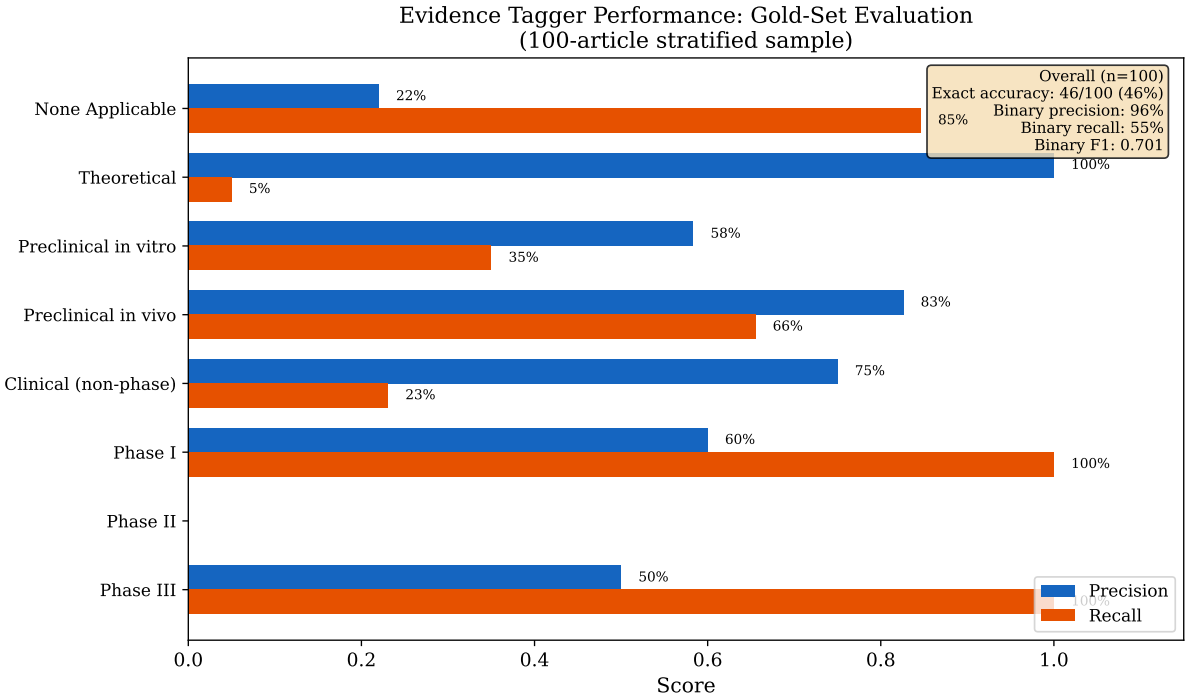

Figure 3.2: Evidence tagger performance: gold-set evaluation (100-article stratified sample).

Short or ambiguous terms may generate false positives, and articles using non-standard terminology may be missed. Second, the archive was constructed from PubMed queries only and does not capture every relevant publication, particularly articles in non-English languages, preprints not yet indexed in PubMed, or studies using unconventional terminology for established mechanisms. Third, the quantitative corpus analyzed here is restricted to the 4,830 records with local full text; although this improves traceability and reproducibility, it also excludes 5,584 abstract-only records that may materially change article-count-based field comparisons. A second-order consequence is that some landmark papers are missing from the local full-text archive even when they are clearly field-defining at the PubMed level; radioligand therapy is the clearest current example because the VISION trial (PMID 34161051) is not locally present. Fourth, evidence-level classification was based on keyword detection plus publication-type metadata (e.g., "Clinical Trial, Phase III", "phase 3", "pilot study", "case report", "in vivo") rather than independent trial registry verification; importantly, evidence-tier coverage remains incomplete even after adding a conservative ‘clinical-other‘ bucket for non-phase patient studies. Claims about the absence of clinical trials for specific mechanisms should therefore be inter-

preted as "not detected in our local keyword-based analysis" rather than definitive statements. Fifth, multi-mechanism co-occurrence counts reflect articles tagged with multiple mechanism keywords, which includes both intentional combination studies and reviews that discuss multiple mechanisms without testing them in combination; a corpus-wide designed-combination classifier (Section 3.13) measures the designed-combination rate at 19.7%, substantially below the 40% co-occurrence rate. Sixth, the 19-mechanism taxonomy has not been tested for sensitivity: collapsing the 7 physical/energy categories into 2-3 broader groups (physical ablation, bioelectric modulation, field-based therapy) would reduce the number of "zero publication gaps" and could change the gap analysis conclusions. Our findings should be interpreted as taxonomy-dependent until sensitivity analyses with alternative groupings are performed. Seventh, our current simulation framework does not fully model in vivo lipid remodeling, including SCD-driven MUFA enrichment and other backup ferroptosis defense programs, so computational support for ferroptosis-based interventions remains stronger for in vitro-like assumptions than for established human tumor biology. Finally, article counts serve as a proxy for research intensity but do not directly measure quality, reproducibility, or clinical significance, and the corpus inherits any publication biases present in the indexed literature.

3.7 Corpus Overview

The final full-text corpus comprised 4,830 articles from 803 journals, with impact metrics available for 3,739 articles via iCite. Publication volume increased 31-fold over the study period, from 38 full-text articles in 2015 to 1,167 in 2025, with 186 articles already indexed in the first quarter of 2026. Growth was non-linear, with a marked inflection point in 2020 (257 full-text articles, a 2.4-fold increase over 2019's 106 articles), coinciding with the maturation of checkpoint immunotherapy, cell therapy, and RNA-platform attention. One important revision-era lesson is that not every post-2020 rise should be read as cancer-therapy-specific signal: for example, mRNA-vaccine counts were materially inflated until supportive and infectious-vaccine spillover was removed from that tag.

The fastest-growing mechanisms by absolute article count between 2020 and 2025 were immunotherapy (+504% annual articles), nanoparticle delivery (+800%), CAR-T therapy (+423%), and sonodynamic therapy (+355%). These growth rates reflect both the maturation of established approaches and the emergence of new platform technologies.

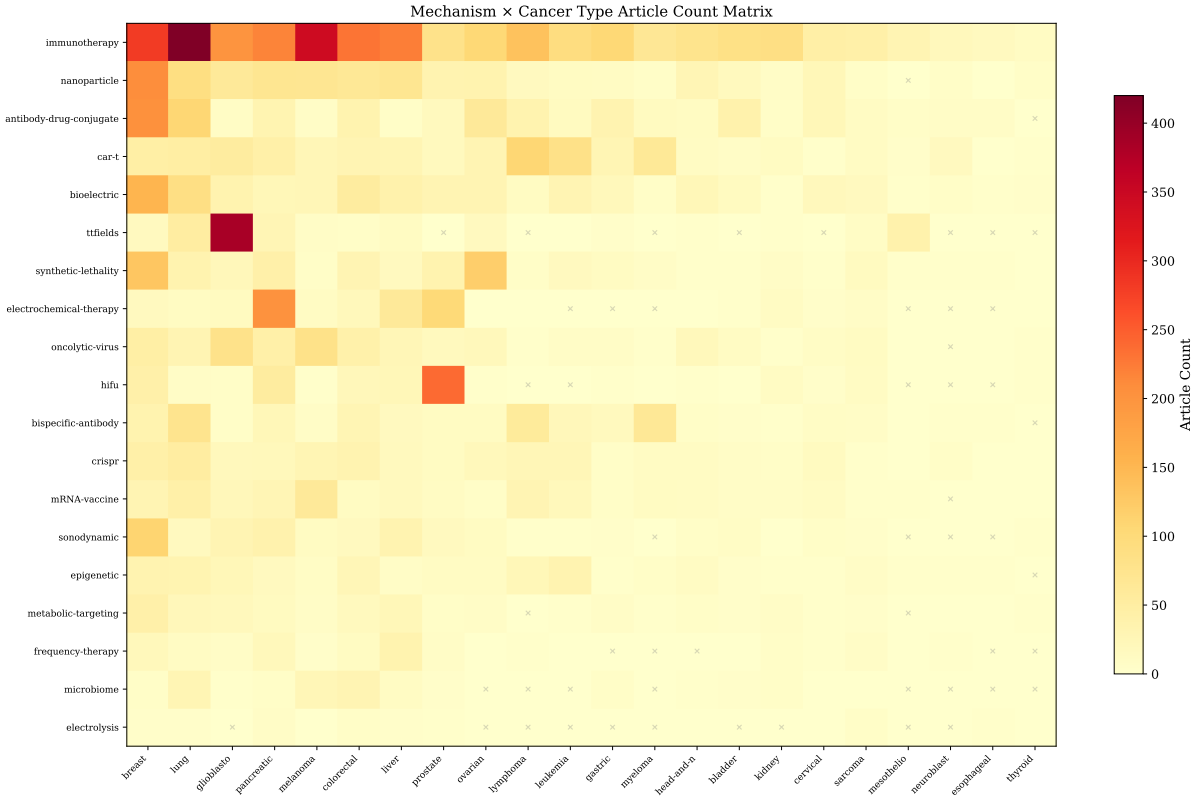

Figure 3.3: Mechanism × cancer type matrix.

3.8 Physical and Energy-Based Mechanisms

3.8.1 Tumor-Treating Fields (TTFields)

TTFields, which deliver low-intensity (1-3 V/cm), intermediate-frequency (100-300 kHz) alternating electric fields to disrupt mitotic spindle formation and organelle assembly in dividing cells O et al. [2021], represent the most clinically advanced physical therapy mechanism in our corpus. We identified 262 TTFields-tagged full-text articles across 14 of 22 cancer types. The evidence-level analysis identified 19 articles classified at the Phase III tier; however, this count still includes secondary analyses and trial-adjacent reports in addition to primary Phase III trial publications. The true number of distinct Phase III TTFields trial reports is therefore smaller than the keyword-derived total.

The evidence base is anchored by the EF-14 trial, a pivotal Phase III randomized clinical trial demonstrating that TTFields plus maintenance temozolomide significantly improved median overall survival (20.9 vs. 16.0 months, $p < 0.001$) and 5-year survival (13% vs. 5%) in newly diagnosed glioblastoma R et al. [2017]. This trial, with 2,494 citations, is the fifth-highest-impact article in our entire corpus (RCR 69.5, 100th percentile). An earlier randomized trial R et al. [2015] (1,203 citations) established the foundation for FDA approval of the Optune device.

Beyond glioblastoma, TTFields are being investigated in mesothelioma (38 articles),

lung cancer (51 articles), pancreatic cancer (27 articles), and ovarian cancer (15 articles). The LUNAR Phase III trial demonstrated improved overall survival in metastatic non-small-cell lung cancer (NSCLC) when TTFields were added to standard systemic therapy T et al. [2023]. A comprehensive review characterizes TTFields as a distinct fourth modality of cancer treatment alongside surgery, radiation, and pharmacotherapy O et al. [2021].

TTFields have zero or near-zero publications in prostate cancer, lymphoma, myeloma, bladder cancer, and cervical cancer in our corpus, though some of these gaps may reflect search query design rather than genuine absence.

3.8.2 High-Intensity Focused Ultrasound (HIFU)

HIFU uses convergent acoustic energy to achieve thermal ablation of tumor tissue at a defined focal point without incision. Our full-text corpus contains 81 HIFU articles across 12 cancer types. The dominant application is prostate cancer (29 full-text articles), where HIFU is established as a focal therapy option E et al. [2015]; S et al. [2018]. Notably, a multicentric study reported 5-year outcomes of focal HIFU for clinically significant nonmetastatic prostate cancer S et al. [2018] (309 citations), demonstrating favorable oncologic control with preserved quality of life.

HIFU research in breast cancer (42 articles), pancreatic cancer (50 articles), and liver cancer (22 articles) is expanding. A systematic review positions HIFU alongside radiofrequency ablation, microwave ablation, and cryoablation as viable alternatives to surgical resection for small tumors van dVEMF et al. [2021].

3.8.3 Sonodynamic Therapy

Sonodynamic therapy (SDT) exploits low-intensity ultrasound to activate sonosensitizer molecules that generate reactive oxygen species (ROS) within tumor tissue. Our full-text corpus contains 187 SDT articles across 15 cancer types, making it one of the most active physical modalities in the energy-based subset. However, SDT faces significant translational challenges. Our keyword-based evidence classification detected only 1 Phase I article in the local full-text corpus, with the balance of the literature remaining heavily preclinical. Broader PubMed verification suggests more early clinical activity may exist outside the locally indexed full-text set, but within the reproducible corpus analyzed here SDT remains overwhelmingly a preclinical field with no detected Phase II or Phase III full-text signal.

The preclinical literature is large and growing. Landmark studies have demonstrated nanoenzyme-augmented SDT with catalytic tumor oxygenation P et al. [2018] (571 citations), graphene-augmented sonocatalytic tumor eradication C et al. [2017] (300 citations), and piezotronic effect-augmented sonosensitizers Y et al. [2022a] (218 citations).

The convergence of SDT with nanoparticle delivery and immunotherapy is particularly active, with emerging evidence for SDT-based immunogenic cell death M et al. [2021a]; Y et al. [2023].

A distinctive molecular signature emerges from our full-text corpus analysis that keeps SDT relevant to the ferroptosis conversation even after broadening the manuscript frame. SDT is strongly associated with **ferroptosis** language — iron-dependent, lipid peroxidation-mediated cell death — with 36 full-text articles linking SDT to ferroptosis, compared to 0 for TTFields, 1 for HIFU, and 3 for IRE in the frozen corpus. More specifically, 79 SDT full-text articles (42% of the SDT corpus) mention the glutathione/GPX4 axis, underscoring how centrally redox buffering appears in this literature. The proposed causal chain — ultrasound-activated ROS generation → glutathione depletion → GPX4 inactivation → lipid peroxidation → ferroptosis → DAMP release → immunogenic cell death — is supported at each individual link by published preclinical data Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; Q et al. [2018]. What this pattern does **not** establish is clinical distinctiveness over PDT; that comparison is addressed in the Discussion.

The primary barriers to clinical translation include the lack of standardized sonosensitizer formulations approved for human use, the absence of validated ultrasound dosimetry protocols for SDT (as distinct from diagnostic or HIFU applications), and the challenge of demonstrating SDT superiority or complementarity to established modalities in registrational trials. Breast cancer is the most studied SDT target in the full-text corpus (42 articles), followed by liver cancer (19 articles) and pancreatic cancer (12 articles).

3.8.4 Frequency Therapy and Electrolysis

Frequency therapy — encompassing pulsed electromagnetic fields (PEMF), radiofrequency ablation in immunomodulatory contexts, and resonance-based approaches — is represented by 71 full-text articles across 11 cancer types. This is a nascent field with mechanistic diversity that complicates systematic evaluation. A foundational review M et al. [2016a] (113 citations) described mechanisms of PEMF in oncology, including disruption of mitotic spindle formation, induction of apoptosis, and immunomodulatory effects.

Electrolysis as a direct antitumor mechanism is the smallest category in our full-text corpus (11 articles across only a handful of cancer contexts), but includes innovative work on amorphous liquid metal electrodes for conformable electrochemical tumor therapy X et al. [2017] (114 citations) and platinum nanoparticle-enabled electrodynamic therapy T et al. [2019] (211 citations).

3.8.5 Electrochemical Therapy

Electrochemical therapy (EChT), dominated by irreversible electroporation (IRE), is represented by 185 full-text articles across 14 cancer types. Pancreatic cancer (61 articles) and prostate cancer (30 articles) are the primary applications. A foundational paper introduced the concept of tissue ablation through irreversible electroporation RV et al. [2005] (1,241 citations), and subsequent clinical development has demonstrated safety and feasibility in locally advanced pancreatic cancer RCG et al. [2015] (400 citations) and localized prostate cancer MJV et al. [2016]; MJ et al. [2018].

A pivotal development is the combination of IRE with immunotherapy: high-frequency IRE has been shown to induce immunogenic cell death Ringel-Scaia et al. [2019] (168 citations), and emerging data demonstrate synergy between IRE and checkpoint blockade plus TLR7 stimulation JSS et al. [2019] (111 citations). The IRE-immunotherapy convergence (45 articles) represents a growing axis of investigation. However, lymphoma, leukemia, gastric cancer, and myeloma have zero EChT publications despite established clinical experience in other solid tumors.

3.9 Bioelectric Modulation

Bioelectric research encompasses membrane potential manipulation, ion channel modulation, voltage-gated signaling in cancer biology, and electroporation-based drug delivery. Our full-text corpus contains 182 articles across 19 cancer types, with broad conceptual range but limited direct clinical antitumor translation.

A landmark Cell paper established the conceptual framework for bioelectric signaling as a reprogrammable circuit underlying embryogenesis, regeneration, and cancer M et al. [2021b] (463 citations), arguing that endogenous bioelectric patterns constitute an informational layer that can be therapeutically manipulated. Earlier work on endogenous voltage gradients controlling growth and form M et al. [2017] (289 citations) provided the biophysical foundation. The identification that mitochondrial membrane potential marks cells with enhanced stemness M et al. [2016b] (370 citations) has implications for targeting cancer stem cell populations.

Despite this conceptual richness, translational progress toward direct antitumor bioelectric interventions has been limited. In the current evidence-tier rerun, no phase-labeled trial evidence was detected for direct bioelectric antitumor therapy. The practical implication is that the bioelectric literature remains conceptually rich but clinically immature within this corpus.

3.10 Immune-Based Approaches

3.10.1 Checkpoint Immunotherapy

Checkpoint immunotherapy is the dominant mechanism in our full-text corpus, with 2,297 articles across all 22 cancer types. This reflects its established clinical role across multiple tumor types. The highest concentrations are in lung cancer (250 articles), melanoma (207 articles), breast cancer (162 articles), pancreatic cancer (138 articles), colorectal cancer (136 articles), liver cancer (133 articles), and glioblastoma (117 articles).

Key landmark articles include a comprehensive review of PD-1/PD-L1 and CTLA-4 checkpoint inhibitors Y et al. [2022b] (1,191 citations), the comprehensive management review establishing temozolomide plus TTFields as the standard of care in glioblastoma and surveying immunotherapy as a promising but significantly challenged area due to the blood-brain barrier and unique immune microenvironment AC et al. [2020] (1,989 citations), and combination immunotherapy for hepatocellular carcinoma reviewing how approximately 30% of HCC exhibits intrinsic resistance to single-agent ICIs and how combination strategies (anti-PD-1/PD-L1 with anti-angiogenics or dual checkpoint blockade) have improved first-line outcomes L et al. [2023] (310 citations). STAT3 targeting as a convergence point for oncogenic signaling and immunomodulation has been extensively reviewed S et al. [2020a] (1,011 citations).

The most active convergence axes for immunotherapy in the full-text corpus include combinations with CAR-T (417 articles), oncolytic viruses (262 articles), nanoparticle delivery (193 articles), mRNA vaccines (131 articles), CRISPR (166 articles), bispecific antibodies (139 articles), metabolic targeting (127 articles), microbiome modulation (100 articles), ADCs (78 articles), SDT (53 articles), and TTFields (47 articles). Immunotherapy remains the central hub of the convergence network.

3.10.2 CAR-T Cell Therapy

CAR-T therapy is represented by 474 full-text articles across all 22 cancer types. The literature has evolved from early hematologic successes in B-cell leukemia and lymphoma S et al. [2020b] (1,400 citations) toward aggressive expansion into solid tumors.

Key developments include: IL-15-armored GPC3-targeted CAR-T cells for solid cancers D et al. [2025] (170 citations), representing a next-generation approach to overcoming the immunosuppressive tumor microenvironment; Claudin18.2-specific CAR-T cells showing promising Phase I results in gastrointestinal cancers C et al. [2022] (546 citations); and IL-10-expressing CAR-T cells that resist dysfunction and mediate durable clearance of solid tumors Y et al. [2024a] (228 citations), demonstrating that cytokine engineering of CAR-T cells can overcome exhaustion.

CRISPR-based engineering of CAR-T cells is a rapidly growing convergence axis (95

full-text articles), exemplified by RASA2 ablation enhancing antigen sensitivity J et al. [2022a] (207 citations) and adenosine A2A receptor deletion enhancing efficacy L et al. [2021] (199 citations). The allogeneic CRISPR-Cas9-engineered CAR-T therapy CTX130, targeting CD70 in T-cell lymphoma, has demonstrated safety in Phase I SP et al. [2025] (58 citations), representing a milestone for gene-edited cell therapies. CAR-T combined with oncolytic viruses (40 articles) and bispecific antibodies (30 articles) represents additional emerging convergence zones.

3.10.3 Bispecific Antibodies

Bispecific antibodies, engineered to simultaneously bind two distinct targets (typically a tumor antigen and an immune effector molecule), are represented by 247 full-text articles across 20 of 22 cancer types. Myeloma is a particularly active area (22 articles), with real-world data on teclistamab demonstrating clinical utility in relapsed/refractory disease C et al. [2024] (94 citations). Lung cancer applications are expanding, with KN046, a PD-L1/CTLA-4 bispecific, showing Phase II results in combination with chemotherapy for first-line metastatic NSCLC Y et al. [2024b] (38 citations), and the EGFRxHER3 bispecific izalontamab reporting first-in-human results J et al. [2025].

Bispecific antibody convergence with ADCs (42 articles) is an active area combining target engagement with cytotoxic payload delivery.

3.10.4 mRNA Cancer Vaccines

mRNA cancer vaccines represent one of the more dynamic but also more taxonomy-sensitive areas in the full-text corpus: 179 articles across 20 cancer types after tightening the mechanism tag to therapeutic cancer-vaccine studies. Earlier drafts materially overcounted this lane because supportive and infectious-vaccine papers in cancer cohorts bled into the mechanism definition. The field was nevertheless catalyzed by the COVID-19 mRNA platform moment, which demonstrated the feasibility of rapid, scalable mRNA production and lipid nanoparticle delivery GT et al. [2022] (279 citations). A comprehensive Lancet Oncology review established the clinical landscape CL et al. [2022] (470 citations), documenting the depth of ongoing therapeutic development.

The landmark article in this category is the Nature report on personalized RNA neoantigen vaccines stimulating T-cell responses in pancreatic ductal adenocarcinoma LA et al. [2023] (1,153 citations, RCR 90.3, 100th percentile) —the second-highest-impact article in our entire corpus. This Phase I trial of autogene cevumeran, an individualized neoantigen vaccine based on uridine mRNA-lipoplex nanoparticles, demonstrated that personalized vaccination could elicit measurable T-cell responses against patient-specific neoantigens in one of the most treatment-resistant cancers.

The mRNA vaccine field still intersects strongly with immunotherapy (131 co-tagged full-text articles) and nanoparticle delivery (47 co-tagged articles), reflecting the inherent dependence of mRNA vaccines on both immune activation and delivery technology. In the current evidence-tier table, the mechanism reaches Phase II, but its local full-text signal remains review-heavy and only partially covered by patient-study tagging. That means the field is clearly active, but mechanistic article counts should not be mistaken for mature comparative clinical evidence. A recent clinical advances review documents the current pipeline across melanoma, glioblastoma, lung, breast, and prostate cancers AV et al. [2025] (74 citations). Lipid nanoparticle-mediated lymph node-targeting delivery of mRNA cancer vaccines has been shown to elicit robust CD8+ T-cell responses J et al. [2022b] (371 citations).

3.10.5 Oncolytic Viruses

Oncolytic virotherapy is represented by 378 full-text articles across 19 cancer types. Glioblastoma is the most studied target (43 articles), followed by melanoma (51 articles), where T-VEC (talimogene laherparepvec) established the precedent for oncolytic virus approval. A comprehensive Nature Reviews Clinical Oncology review characterizes the field's progress and challenges SZ et al. [2023] (489 citations), noting that oncolytic viruses offer selective replication in tumor cells, delivery of multiple transgene payloads, induction of immunogenic cell death, and promotion of antitumor immunity.

A notable 2025 Cell paper reported a hyperacute rejection-engineered oncolytic virus designed for interventional clinical trials in refractory cancer patients L et al. [2025] (67 citations), addressing the critical limitations of intravenous safety and inherent immune neutralization. In hepatocellular carcinoma, the oncolytic virus VG161 showed activity in a Nature-published Phase I study Y et al. [2025] (53 citations). The emerging field of OV-based cancer immunotherapy has been comprehensively reviewed in Trends in Cancer R et al. [2023] (264 citations), positioning oncolytic viruses as multi-faceted tumor killers that combine direct lysis with immune potentiation.

The strongest convergence axis is oncolytic virus-immunotherapy (262 full-text articles), consistent with the mechanistic rationale that viral oncolysis generates immunogenic cell death that may synergize with checkpoint blockade SZ et al. [2023]; R et al. [2023]. Combinations with CAR-T (40 articles) and nanoparticle delivery remain emergent.

3.11 Precision Molecular Approaches

3.11.1 CRISPR Gene Editing

CRISPR-based approaches are represented by 331 full-text articles across 20 cancer types, with no Phase II or Phase III articles detected by our automated classification, placing CRISPR among the most preclinical-heavy mechanisms in our analysis. Despite this, the conceptual impact is enormous: CRISPR screens have defined unified hallmarks of cancer cell-autonomous immune evasion, providing a systematic map of therapeutic targets A et al. [2026].

Key convergence axes include CRISPR-immunotherapy (166 articles), CRISPR-CAR-T (95 articles), CRISPR-synthetic lethality (50 articles), and CRISPR-nanoparticle (29 articles). The engineering of next-generation CAR-T cells with CRISPR-Cas9 has been extensively reviewed A et al. [2022] (314 citations), and clinical milestones include non-viral precision T-cell receptor replacement for personalized cell therapy SP et al. [2023] (256 citations) and the COBALT-LYM Phase I trial of CRISPR-edited allogeneic CAR-T for T-cell lymphoma SP et al. [2025] (58 citations).

Precision nanomedicine integrating CRISPR with nanoparticle delivery for tumor microenvironment modulation represents a growing area H et al. [2025a] (101 citations).

3.11.2 Synthetic Lethality

Synthetic lethality — the paradigm in which loss of two genes is lethal while loss of either alone is viable —is represented by 367 full-text articles across 21 cancer types. PARP inhibitors, the clinical embodiment of this concept, have achieved broad clinical use in breast cancer (69 articles), ovarian cancer (73 articles), pancreatic cancer (25 articles), and prostate cancer (24 articles). A pan-tumor review documented the expanding role of PARP inhibitors across multiple malignancies CC et al. [2025] (21 citations).

The concept of synthetic lethality as an engine for cancer drug target discovery has been comprehensively reviewed A et al. [2020] (458 citations), and CRISPR-synthetic lethality combinations (63 articles) represent a powerful approach for systematic identification of new synthetic lethal pairs. Transcription-replication conflicts have been identified as a mechanistic basis for PARP inhibitor sensitivity M et al. [2024] (160 citations), expanding the theoretical framework beyond BRCA mutations.

Our corpus detected zero synthetic lethality-myeloma co-tagged articles; however, independent PubMed verification reveals at least 11 publications on PARP inhibitors in multiple myeloma, including early-phase clinical data on veliparib combined with bortezomib. This gap is therefore an artifact of our search query design rather than a genuine absence of research. The myeloma-PARP axis is under-researched relative to breast or ovarian cancer applications, but it is not unexplored.

3.11.3 Antibody-Drug Conjugates (ADCs)

ADCs are represented by 284 full-text articles across 21 cancer types. The field is dominated by breast cancer applications (92 articles), lung cancer (48 articles), and ovarian cancer (29 articles). The changing treatment landscape of EGFR-mutant NSCLC has been shaped by ADC development alongside bispecific antibodies F et al. [2025] (106 citations).

ADC convergence with immunotherapy (160 articles) is well established, and combinations with CAR-T (42 articles) and bispecific antibodies (42 articles) are emerging.

3.11.4 Epigenetic Therapy

Epigenetic approaches (DNA methylation inhibitors, histone deacetylase inhibitors, bromodomain inhibitors, and related agents) are represented by 183 full-text articles across 20 cancer types. A landmark Cell Death Discovery review established the current state of clinical translation X et al. [2024] (223 citations), and a Signal Transduction and Targeted Therapy paper provided a comprehensive framework for epigenetic regulation in the tumor microenvironment J et al. [2023] (322 citations).

The most impactful single finding in this category is the Nature paper demonstrating that NBS1 lactylation is required for efficient DNA repair and chemotherapy resistance H et al. [2024] (413 citations, RCR 56.2), linking metabolic reprogramming (the Warburg effect) to epigenetic modification of the DNA damage response. This exemplifies how mechanistic convergence at the molecular level can reveal new therapeutic vulnerabilities.

Epigenetic-immunotherapy convergence (78 articles) is the dominant combination.

3.12 Delivery and Microenvironment

3.12.1 Nanoparticle Delivery

Nanoparticle-based delivery systems constitute the second-largest mechanism category in our full-text corpus: 515 articles across 20 cancer types. An important distinction must be drawn between established and experimental nanoplatforms. Several nanomedicines have achieved FDA approval, including liposomal doxorubicin (Doxil, 1995), nab-paclitaxel (Abraxane, 2005), liposomal irinotecan (Onivyde, 2015), and liposomal daunorubicin-cytarabine (Vyxeos, 2017) MD et al. [2023]. These approved nanomedicines are typically described by drug name rather than "nanoparticle" in the literature and are therefore underrepresented in our corpus. The translational gap identified here primarily applies to novel experimental nanoplatforms — metal, polymer, and inorganic nanoparticles — which remain overwhelmingly preclinical in our full-text corpus. This distinction between proven liposomal/albumin-bound formulations and the broader experimental

nanoparticle literature is critical: the former demonstrate that nanomedicine can succeed clinically, while the latter face substantial barriers to replicating that success.

The nanoparticle literature spans gold, silver, and iron oxide nanoparticles P et al. [2025] (88 citations), liposomal formulations MD et al. [2023] (215 citations), lipid nanoparticles for mRNA delivery J et al. [2022b] (371 citations), and precision nanomedicine for tumor microenvironment modulation H et al. [2025a] (101 citations). The convergence with immunotherapy (193 articles) and sonodynamic therapy (116 articles) is extensive, as nanoparticles serve as both therapeutic agents and delivery vehicles for other modalities.

The fundamental translational barriers include: (a) heterogeneous nanoparticle pharmacokinetics and biodistribution that are difficult to predict from preclinical models; (b) manufacturing scalability and reproducibility challenges; (c) regulatory uncertainty around the classification and approval pathway for nanoparticle-drug combinations; and (d) the enhanced permeability and retention (EPR) effect, which is robust in rodent tumors but inconsistent in human cancers. Breast cancer (172 articles), lung cancer (76 articles), and liver cancer (66 articles) are the primary targets. Mesothelioma has zero nanoparticle publications despite the appeal of local delivery to pleural surfaces.

3.12.2 Metabolic Targeting

Metabolic targeting — exploitation of altered cancer cell metabolism, particularly the Warburg effect, glutamine addiction, and lipid metabolism dependencies —is represented by 274 full-text articles across 20 cancer types. A comprehensive Nature Reviews Drug Discovery article ZE et al. [2022] (1,127 citations) chronicles the hundred-year journey from Warburg's discovery to contemporary precision metabolic oncology.

The convergence of metabolic targeting with immunotherapy (127 articles) reflects the recognition that metabolic reprogramming of the tumor microenvironment suppresses immune function and that metabolic interventions can restore immunocompetence. The landmark demonstration that glycolysis modulation can enhance checkpoint immunotherapy C et al. [2023] (341 citations) illustrates this principle. Glutamine addiction as a therapeutic target was comprehensively reviewed in a highly cited Trends in Biochemical Sciences paper DR et al. [2010] (1,747 citations).

Metabolic targeting has limited representation in lymphoma within our corpus, though broader PubMed searches suggest this gap partly reflects our search query design.

3.12.3 Microbiome Modulation

Microbiome modulation is the most recently emergent mechanism in our analysis: 109 full-text articles across 11 cancer types. The field has been catalyzed by the recognition that gut microbiota composition influences checkpoint immunotherapy response, as

comprehensively reviewed in Nature Reviews Cancer MR et al. [2022] (258 citations) and Nature Medicine EM et al. [2022] (467 citations).

Fecal microbiota transplantation (FMT) combined with checkpoint inhibitors has emerged as a clinical strategy, with a Gut Microbes review characterizing optimization approaches A et al. [2025a] (56 citations) and a meta-analysis evaluating FMT-ICI combinations A et al. [2025b] (23 citations). The DIET randomized trial investigated dietary intervention to modulate microbiome composition in melanoma patients receiving immunotherapy RM et al. [2024] (21 citations).

Microbiome-immunotherapy convergence (100 articles) dominates the microbiome literature in our corpus.

3.13 Cross-Mechanism Convergence: Where Synergies Emerge

Of the 4,830 articles in our full-text corpus, 1,952 (40%) are tagged with two or more therapeutic mechanism keywords, and 396 carry three or more tags. To distinguish deliberate combination studies from incidental co-mentions, we applied a corpus-wide designed-combination classifier using explicit combination/synergy language and study-type signals. Of the 1,952 multi-mechanism articles, 385 (19.7%) are classified as designed combination studies — 60 with clinical patient-study signal and 325 preclinical — while 622 (31.9%) are co-mentions only and 945 (48.4%) are reviews or perspectives spanning multiple lanes (Figure 3.4). An earlier 50-article manual subsample had estimated ∼10% experimental combinations; the automated classifier yields a higher rate primarily because it captures preclinical combination designs that the small manual sample underrepresented. The remaining co-tagged articles map the intellectual territory where researchers are connecting mechanisms conceptually — a precursor to deliberate combination design, but not equivalent to experimental evidence of synergy. We distinguish between the co-occurrence rate (40%) and the measured designed-combination rate (19.7%), and caution against conflating the two.

The convergence landscape is organized around a central hub: checkpoint immunotherapy participates in 8 of the top 10 mechanism pairs and co-occurs with every other mechanism in the corpus. The top 10 convergence pairs are:

1. **CAR-T + Immunotherapy:** 417 articles. This dominant axis reflects the recognition that CAR-T cells face significant barriers in solid tumors — including exhaustion, antigen escape, and immunosuppressive microenvironments — and that checkpoint blockade can partially rescue exhausted CAR-T cell function S et al. [2020b]. 2. **Immunotherapy + Oncolytic Virus:** 262 articles. Oncolytic viruses induce immunogenic cell death that synergizes with checkpoint blockade by converting immunologically "cold"

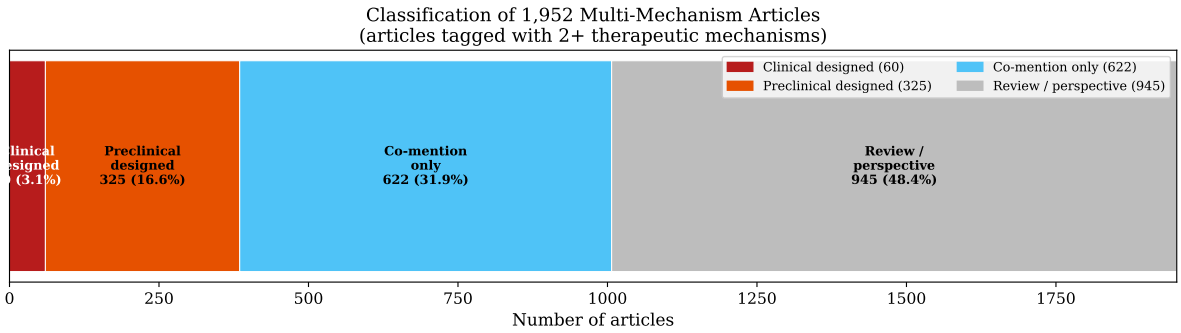

Figure 3.4: Classification of multi-mechanism articles into designed combinations, co-mentions, and reviews.

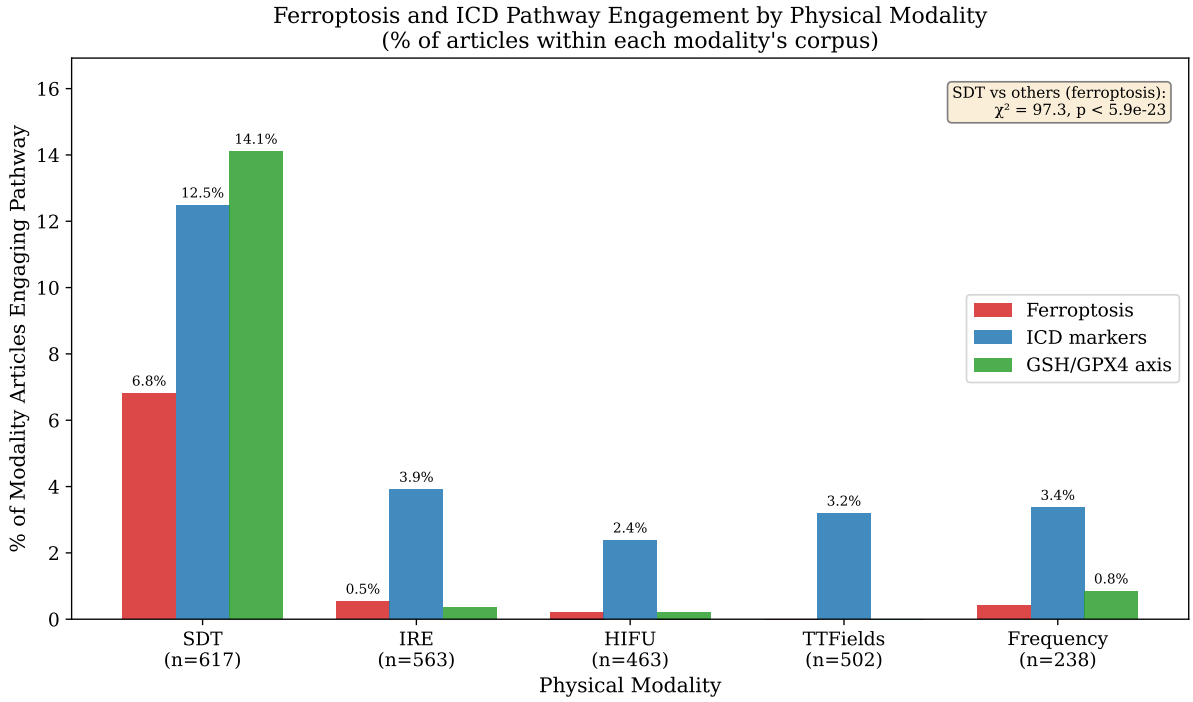

Figure 3.5: Ferroptosis engagement ($\chi^2 = 97.3$, $p < 5.9 \times 10^{-23}$).

tumors to "hot." 3. **Immunotherapy + Nanoparticle:** 193 articles. Nanoparticles serve as delivery vehicles for checkpoint inhibitor payloads, immune adjuvants, and tumor antigens for in situ vaccination. 4. **Immunotherapy + mRNA Vaccine:** 131 articles. mRNA vaccines generate neoantigen-specific T-cell responses that are amplified by checkpoint blockade. 5. **CRISPR + Immunotherapy:** 166 articles. CRISPR screens identify immune evasion mechanisms; CRISPR editing engineers superior immune cells. 6. **Bispecific Antibody + Immunotherapy:** 139 articles. Bispecific antibodies redirect immune effectors to tumors while checkpoint inhibitors release immune brakes. 7. **Immunotherapy + Metabolic Targeting:** 127 articles. Metabolic interventions can reshape the tumor microenvironment to restore immune function and improve checkpoint responses. 8. **Nanoparticle + Sonodynamic:** 116 articles. Nanoparticles function as

sonosensitizer carriers, enhancing SDT specificity and efficacy through targeted delivery and tumor microenvironment responsiveness. 9. **Immunotherapy + Microbiome:** 100 articles. Gut microbiota composition modulates systemic immune competence and checkpoint inhibitor efficacy. 10. **Bioelectric + Nanoparticle:** 34 articles. Nanoparticles enable bioelectric modulation through electroporation-mediated delivery and conductive nanoparticle-enhanced electric field effects.

Notable triple-mechanism articles include the landmark thermal ablation review integrating electrochemical therapy, frequency therapy, and immunotherapy KF et al. [2014] (1,938 citations), the personalized mRNA neoantigen vaccine study combining immunotherapy, mRNA vaccine, and nanoparticle platforms LA et al. [2023] (1,153 citations), and the CRISPR-engineered CAR-T review A et al. [2022] (314 citations).

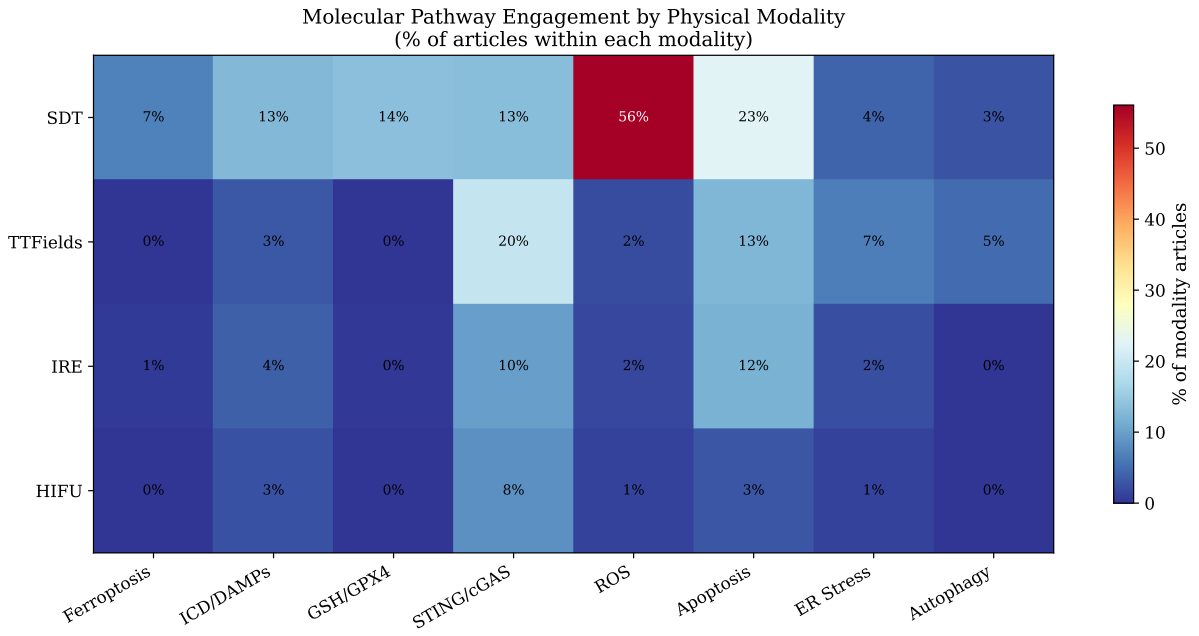

Figure 3.6: Molecular pathway engagement (normalized %).

3.14 The Mechanism-Cancer Matrix: What Works Where

The 19 x 22 mechanism-cancer matrix reveals both deep concentrations and systematic gaps.

Dominant mechanism-cancer pairs (top 5 by article count): immunotherapy-lung (250), immunotherapy-melanoma (207), TTFields-glioblastoma (204), immunotherapy-breast (162), and immunotherapy-pancreatic (138). These represent the densest full-text research axes in the current indexed corpus, though maturity of evidence varies substantially across them.

Broad-spectrum mechanisms —those studied across the widest range of cancer

types —include CAR-T therapy and checkpoint immunotherapy (22/22 cancer types), synthetic lethality and ADCs (21/22), and CRISPR, bispecific antibodies, epigenetic therapy, mRNA vaccines, metabolic targeting, and nanoparticles (20/22).

Narrow-spectrum mechanisms include electrolysis (3/22), microbiome modulation and frequency therapy (11/22 each), HIFU (12/22), TTFields and electrochemical therapy (14/22 each), and sonodynamic therapy (15/22). The narrowness of TTFields coverage is particularly noteworthy given its strong clinical evidence base.

The most intensively researched cancer types across all mechanisms are breast cancer (697 total cross-mechanism article-instances), lung cancer (593), glioblastoma (529), pancreatic cancer (428), melanoma (425), and colorectal cancer (376). The least-researched cancer types are thyroid (19), esophageal (20), mesothelioma (49), and neuroblastoma (52).

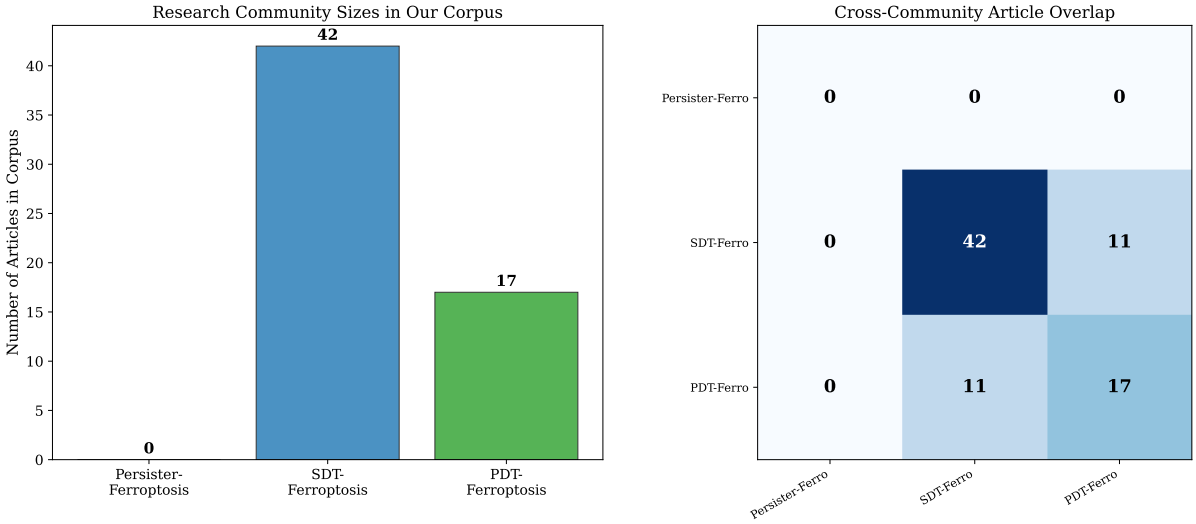

Figure 3.7: Literature disconnect between communities.

Chapter 4

Evidence Quality and Gaps

4.1 Evidence Maturity and Weighted Scoring

Evidence tags are currently populated for 2,038 of 4,830 full-text records (42.2%). Reviews and meta-analyses (1,921) and protocols (29) are intentionally left unclassified; among the remaining primary-study-like records, coverage is 2,038 of 2,880 (70.8%). The gold-set evaluation described in Section 3.6 confirms that this coverage reflects a conservative detector: tagged evidence is almost always real, but a substantial share of valid evidence goes undetected, particularly in the clinical-other, theoretical, and preclinical-in-vitro categories.

To move beyond raw counts, we computed a weighted evidence score for each mechanism that combines evidence tier (Phase III = 12 points, down to theoretical = 0.5), citation percentile (1.0-1.5x modifier), and publication recency (0.9-1.1x modifier). The resulting ranking shows immunotherapy (2,617), bispecific antibodies (687), TTFields (568), and ADCs (558) as the most evidence-mature mechanisms. Weighted scoring highlights where article volume is not matched by evidence depth: nanoparticles rank sixth by weighted evidence despite ranking second by article count, reflecting their overwhelmingly preclinical composition. Bioelectric modulation has the lowest average weight per tagged article (1.69), despite 144 tagged evidence rows, indicating high volume but minimal clinical signal.

4.2 Main Contribution: A Broader Frame Before A Narrower Hypothesis

The most important contribution of this article is not a claim that one modality should replace another. It is a change in framing. The frozen 19-mechanism corpus shows that oncology is not short of therapeutic ideas; it is short of disciplined ways to compare them. Mechanism counts, convergence maps, and evidence tiers reveal a field dominated

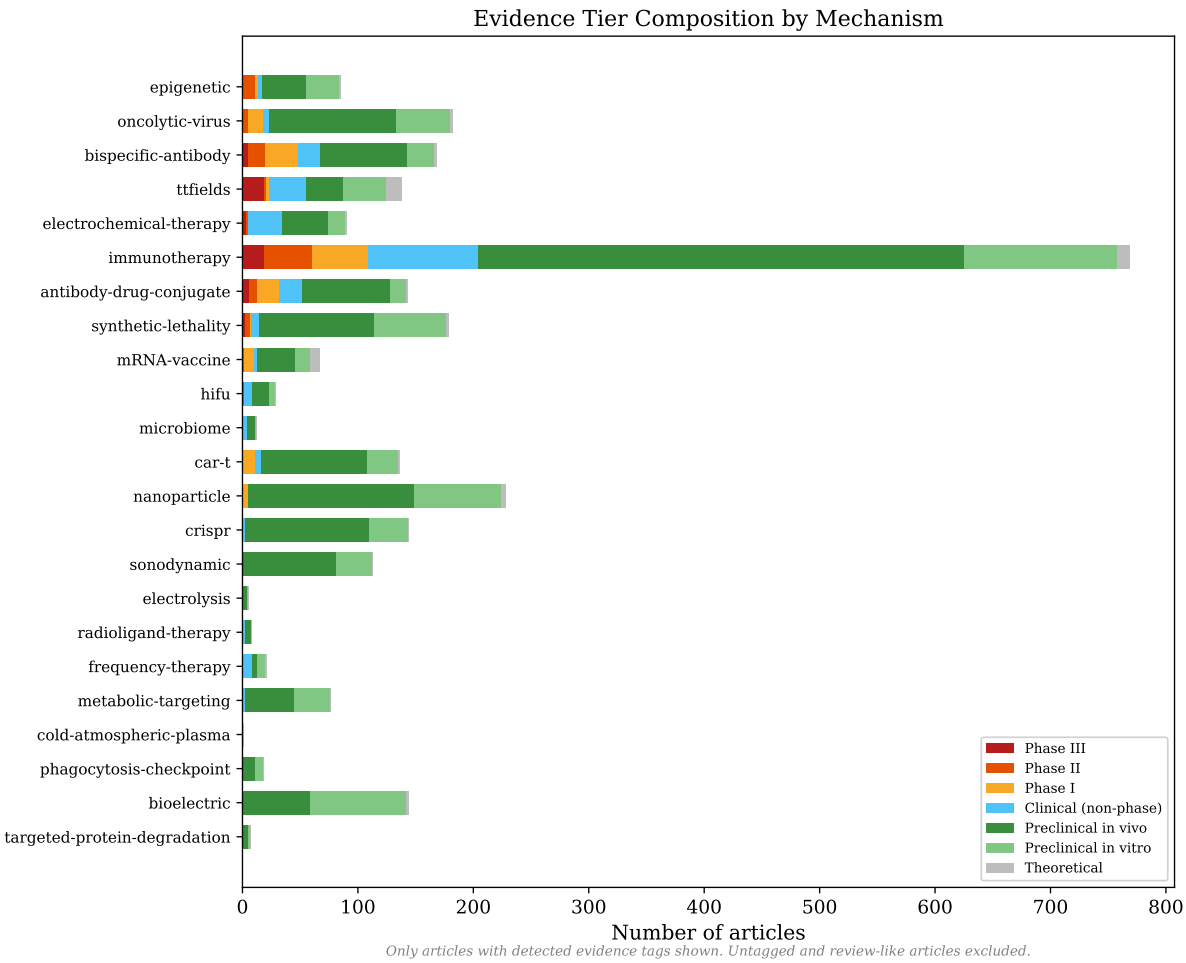

Figure 4.1: Evidence tier composition by mechanism.

by a few high-volume hubs, but they also show how quickly apparently clean conclusions can be destabilized by taxonomy choices, missing full text, incomplete evidence tagging, and tag contamination from adjacent literatures. That matters because many attractive "gaps" are not true absences, and many active literatures are not equally mature.

Within that broader frame, one narrower implication remains worth pursuing. If residual disease states are genuinely OXPHOS-skewed, lipid-peroxidation-prone, and locally targetable, then physical ROS-generating modalities may be worth evaluating alongside pharmacologic ferroptosis inducers rather than outside the conversation. That is a conditional hypothesis downstream of the landscape analysis, not the article's starting assumption.

A tissue-of-origin layer derived from existing cancer-type tags provides a coarser but more interpretable view of where research is concentrated. Of the 4,830 full-text records, 2,995 unique articles (62.0%) have at least one derived tissue category. Because a single article can study multiple cancer types and therefore map to more than one tissue group, the per-category counts below are overlapping and sum to 3,241 rather than 2,995: epithelial (2,212), neuroectodermal (667), hematologic (245), mesenchymal (77), and mesothelial

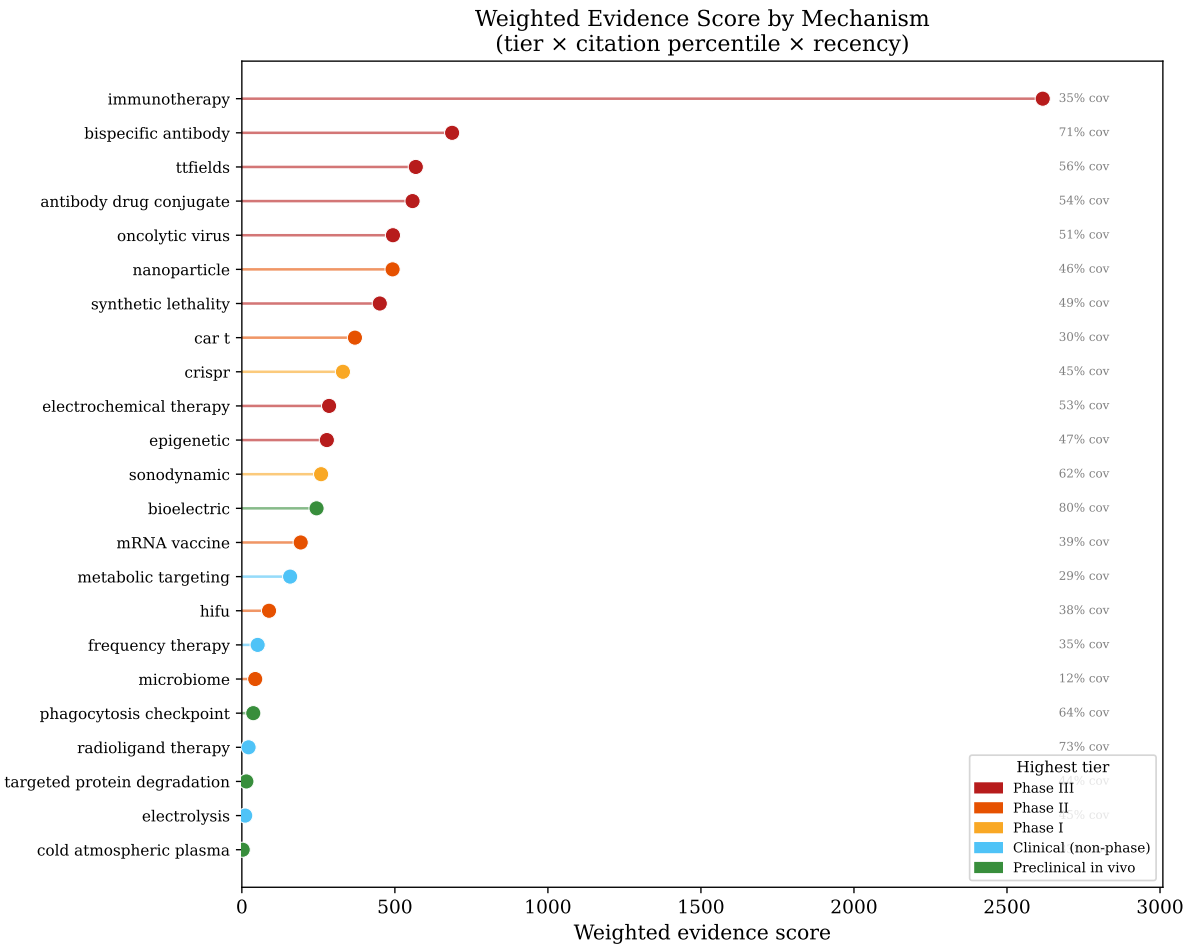

Figure 4.2: Weighted evidence score by mechanism (tier × citation percentile × recency).

(40). The physical-modality literature is overwhelmingly concentrated in epithelial and neuroectodermal contexts, with hematologic and mesothelial categories containing almost no physical-modality signal (Figure 4.3). This serves as a check against over-generalizing localized solid-tumor strategies to all cancers. The tissue layer is derived from the same cancer-type tags used in the mechanism-cancer matrix — it improves interpretation but does not fix upstream tagging errors.

4.3 What The Corpus Actually Supports About PDT, SDT, and CAP

The corpus and external cross-checking support a more modest position than our earlier framing implied.

First, PDT is the established precedent. PDT has the larger ferroptosis and ferroptosis-plus-ICD literature, more mature clinical history, and approved photosensitizers. Any argument for SDT must therefore explain why it adds something beyond a

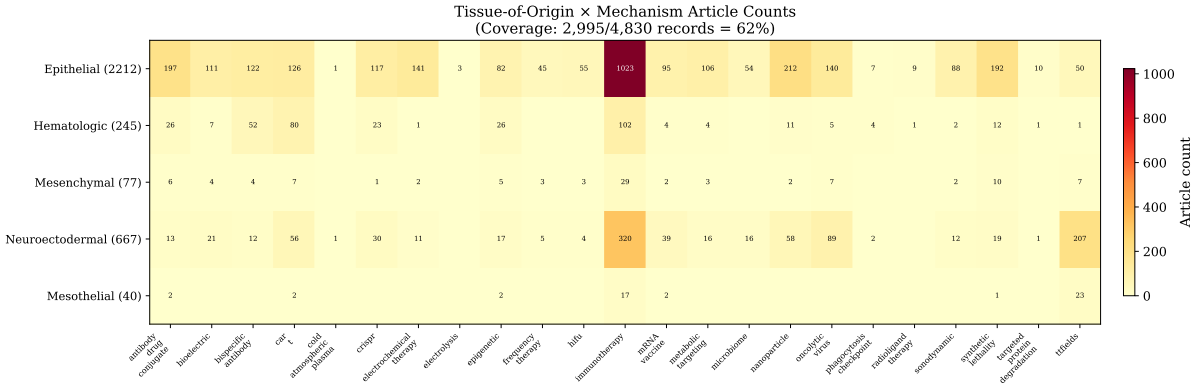

Figure 4.3: Tissue-of-origin × mechanism article counts (coverage: 62%).

depth extension of an older modality rather than pretending SDT is uniquely linked to ferroptosis or ICD.

Second, SDT's real advantage is anatomical, not conceptual. Ultrasound reaches deep soft tissue that light cannot. For pancreatic, hepatic, or pelvic disease, SDT has a genuine physical-access advantage over PDT. But the mechanism remains broadly shared: externally triggered ROS, redox-buffer exhaustion, lipid peroxidation, and potential DAMP release. The clinically relevant question is therefore where depth changes the therapeutic opportunity enough to matter.

Third, CAP is a speculative adjacent lane, not a co-equal conclusion. CAP broadens the class of externally imposed ROS therapies, but in the current local corpus it is represented by only 3 full-text articles and should therefore be treated as a hypothesis-generating comparator for accessible or locoregional settings, not as a mature pillar of the argument.

Fourth, the strongest version of the PDT/SDT claim is now conditional. Physical ROS modalities are most plausible when residual disease is locally targetable, when spatial selectivity matters more than systemic reach, and when ferroptosis-related vulnerability is expected to persist long enough for intervention. The unresolved in vivo problem is exactly where this claim can fail: SCD/MUFA remodeling, DHODH-linked backup defenses, NRF2 compensation, stromal buffering, and other context-dependent escape programs may narrow or erase the apparent window.

Part III

What The Simulations Show

Chapter 5

The Ferroptosis Engine

5.1 Computational Simulation: Ferroptosis Sensitivity Across Cell Phenotypes

To test whether the proposed persister ferroptosis vulnerability is biochemically plausible, we developed a Monte Carlo simulation of the ferroptosis cascade. The model simulates the ROS → GSH depletion → GPX4/FSP1 inactivation → lipid peroxidation → cell death chain for individual cells with stochastic parameter variation, using literature-sourced biochemical values (Methods in Supplementary). We simulated 1,000,000 cells for each of 16 conditions (4 phenotypes × 4 treatments = 16,000,000 total cells).

Model features: (i) autocatalytic lipid peroxidation propagation gated by GSH/GPX4 — creating a bistable switch where cells either recover or collapse irreversibly (Porter et al., Chem Rev 2005); (ii) dynamic GPX4 regulation — degraded by oxidative stress, upregulated by NRF2 (Dodson et al., Free Radic Biol Med 2019); (iii) FSP1 as a GPX4-independent repair pathway (Bersuker et al., Nature 2019), downregulated in the persister phenotype (Higuchi et al., PMID:41481741); (iv) uncapped GSH depletion under exogenous ROS.

Calibration constraint: All phenotypes must show <2% death under Control (no treatment), reflecting the biological reality that even fragile persister cells are viable at baseline.

Results (Figure 7):

Table 5.1: Monte Carlo ferroptosis simulation (n=1M cells/condition).

Phenotype	**Control**	**RSL3**	**SDT**	**PDT**
Glycolytic	0.00%	0.00%	87.2%	87.2%
OXPHOS	0.04%	1.1%	99.9%	99.9%
Persister (FSP1↓)	1.2%	42.5%	100.0%	100.0%
Persister + NRF2	0.00%	0.05%	99.5%	99.5%

Key findings from the simulation:

1. **RSL3 selectively kills persisters (42.5%) while sparing glycolytic cells (0%)**, consistent with experimental findings (PMID:41481741). The model correctly reproduces the published biology: FSP1 downregulation leaves GPX4 as the sole lipid repair pathway, and its pharmacologic inhibition is lethal.

2. **SDT/PDT kills all cell types at high rates (87-100%)** because the exogenous ROS burst overwhelms GSH defense regardless of phenotype. This is an important finding: **physical ROS modalities are not inherently selective for persisters**. Their advantage over RSL3 is not selectivity but spatial confinement (focused ultrasound/light limits damage to the tumor volume). Any ICD advantage remains a biological hypothesis, not a result established by this model.

3. **The bistable switch is the critical mechanism.** Under SDT, GSH is catastrophically depleted (5.0 $\rightarrow$ 0.03 mM), which unlocks autocatalytic lipid peroxidation propagation. Cells that maintain GSH above the threshold recover; those that don't collapse irreversibly. This is qualitatively different from a linear dose-response.

4. **NRF2 compensation fully rescues from RSL3** (0.05% death, matching NRF2's known role in GPX4 upregulation) **but NOT from SDT** (99.5% death). The ROS burst from SDT overwhelms even NRF2-elevated GSH. This suggests that the NRF2 compensation failure mode we identified as the paper's primary risk may be less problematic for SDT than for pharmacologic ferroptosis inducers — a novel prediction from the simulation.

5. **Sensitivity analysis**: The qualitative result tested in the sensitivity suite — Persister > Glycolytic under SDT — holds in 22/22 parameter perturbation conditions ($\pm$50% on each of 11 rate constants), indicating robustness to parameter uncertainty for that specific comparison.

Model limitations: This is a qualitative model testing directional predictions, not a quantitative predictor of clinical death rates. The absolute percentages (87%, 42%, etc.) depend on model-specific rate constants and should not be interpreted as clinical predictions. Its value is in demonstrating that the proposed biochemical chain produces the expected differential sensitivity under biologically plausible parameters. All $\sim$30 simulation parameters are documented with literature sources and confidence ratings. A calibration layer evaluates 5 targets against existing simulation outputs: 3 are independent calibration targets (persister RSL3 death rate, RSL3 window closure timing, MUFA protection factor) and 2 are self-consistency checks on physics parameters (PDT attenuation, SDT penetration).

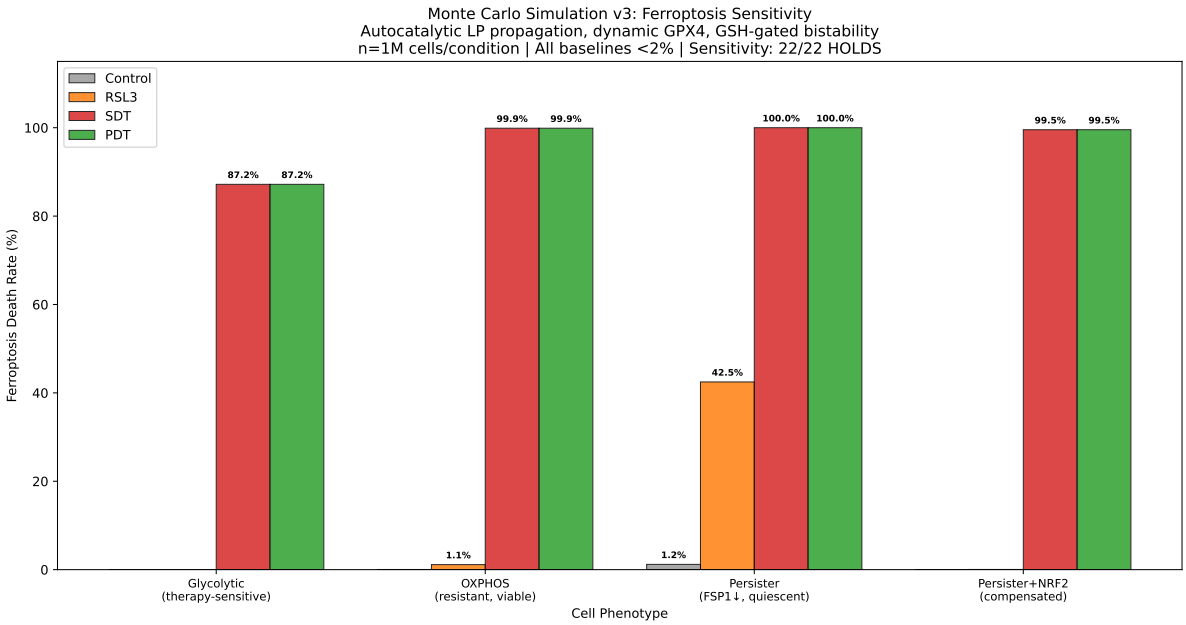

Figure 5.1: Monte Carlo simulation (1M cells/condition).

Chapter 6

Drug Combinations and Penetration

6.1 Spatial Tumor Simulation: Depth-Dependent Energy Deposition

To address the single-cell model's lack of spatial context, we extended the simulation to a 2D heterogeneous tumor grid (500 x 500 cells, 20 um/cell = 1 cm x 1 cm tissue cross-section) with physics-based energy attenuation. The tumor architecture consists of a glycolytic core (80% glycolytic, 15% OXPHOS, 5% persister), an OXPHOS-rich periphery (70% OXPHOS, 20% glycolytic, 8% persister, 2% persister+NRF2), persister cluster pockets, and a stromal border — totaling 159,033 tumor cells per simulation.

Energy is applied from the tissue surface (row 0) with physically grounded attenuation models:

- **PDT**: Modified Beer-Lambert exponential decay with effective attenuation coefficient 0.31 per mm, yielding a penetration depth of approximately 3.2 mm at 630 nm (Jacques, Phys Med Biol 2013) - **SDT**: Acoustic intensity attenuation at 0.7 dB/cm/MHz and 1 MHz operating frequency (Cobbold, Foundations of Biomedical Ultrasound, 2007) - **RSL3**: Uniform concentration (systemic drug, no depth dependence)

Dead cells release labile iron to their Moore neighborhood (8-neighbors), modeling the bystander ferroptosis effect. Iron is consumed by Fenton chemistry each timestep rather than accumulating permanently.

Results (Figure 8):

Key findings from the spatial simulation:

1. **PDT kill rate collapses beyond 3 mm.** At 1 mm depth, PDT kills 90.8% of tumor cells; at 5 mm, only 15.7%; at 9 mm, 0.5%. This directly reflects the exponential light attenuation in tissue and quantifies why PDT is limited to superficial tumors.

2. **SDT maintains >88% kill rate through the full 1 cm depth.** Ultrasound attenuation at 1 MHz is mild (∼0.7 dB/cm), preserving sufficient acoustic intensity to activate sonosensitizers at centimeter depths. This validates the paper's central claim:

SDT extends ROS-ferroptosis to deep-seated tumors where PDT cannot reach.

3. **RSL3 kills uniformly but weakly (∼3-6%).** As a systemic drug, RSL3 shows no depth dependence — it reaches all cells equally. However, it kills selectively: 42.2% of persisters (4,815/11,391) versus 0% of glycolytic cells, consistent with the single-cell model.

4. **Overall tumor kill: SDT 91.9%, PDT 26.1%, RSL3 3.7%, Control 0.1%.** PDT killed only 26.1% of the heterogeneous tumor because 74% of cells lie beyond its effective penetration depth. SDT killed 91.9% — a 3.5-fold advantage over PDT in this 1 cm tissue model.

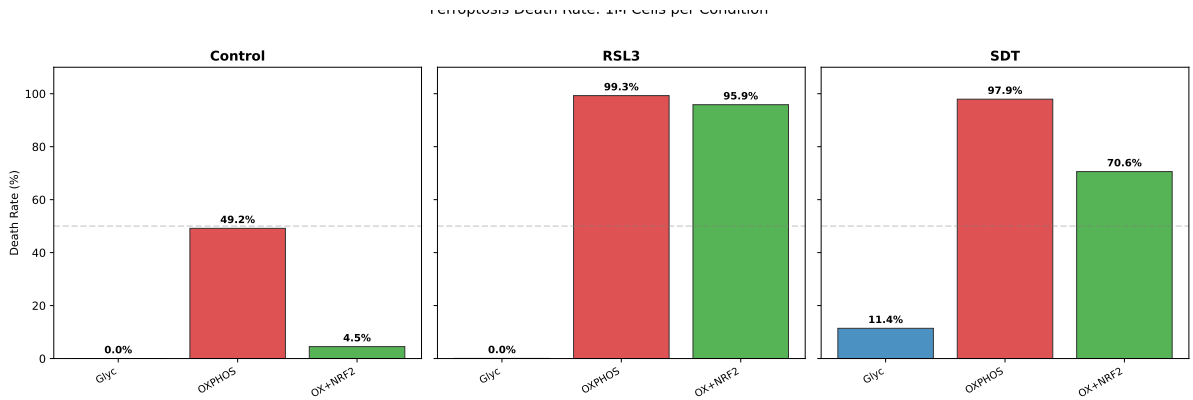

Figure 6.1: Spatial tumor simulation: depth-kill curves and 2D death heatmaps.

The spatial model treats RSL3 as uniformly distributed, but real drug concentration drops with distance from the nearest blood vessel. A tissue-specific drug penetration module using the exponential Krogh cylinder approximation ($C(r) = C_0 \times \exp(-r/\lambda)$, where $\lambda = \sqrt{D/k}$ is the penetration length) shows that even at the vessel wall (the highest-concentration position), tissue-specific vascular permeability reduces effective drug exposure — yielding vessel-wall persister kill rates of 12.1% in well-vascularized epithelial tissue, 2.6% in poorly-vascularized tissue (pancreatic), and 1.8% behind the blood-brain barrier (glioblastoma), with overall tumor kill rates even lower (6.6%, 2.1%, 1.5% respectively). These are substantially lower than the 40% kill rate observed in 2D culture, reinforcing the in-vitro-to-in-vivo translation gap. All transport parameters are estimated, not calibrated — the comparison across tissue types is directionally informative, not quantitatively precise.

6.2 Vulnerability Window: How Long Does the Ferroptosis-Sensitive Window Stay Open?

A critical question for clinical translation is timing: how long after chemotherapy withdrawal do persister cells remain ferroptosis-sensitive? We modeled the recovery of de-

fense pathways (FSP1, GPX4, NRF2, GSH) using exponential kinetics with literature-estimated half-times (FSP1: 7 days, GPX4: 3 days, NRF2: 5 days, GSH: 1 day) and simulated 100,000 cells per condition at each timepoint.

The RSL3 vulnerability window closes within 3 days as GPX4 is re-expressed. By one week, RSL3 is completely ineffective (0.01%). **The SDT window remains open for at least 4 weeks** (99.5% kill at day 28), because the exogenous ROS burst overwhelms even partially recovered antioxidant defenses. This differential has direct clinical implications: pharmacologic ferroptosis inducers require intervention within days of chemotherapy, while physical modalities offer a much wider scheduling window. To our knowledge, this is the first computational estimate of the persister ferroptosis vulnerability window duration.

6.3 Mechanistic Combination Modeling: Why Drug Pairs Synergize

The ferroptosis pathway has two parallel repair arms: GPX4 (GSH-dependent) and FSP1 (CoQ10-dependent). These feed into a shared antioxidant quench term that suppresses autocatalytic lipid peroxidation propagation. Depleting both arms simultaneously should produce mechanistic synergy — the simulation lets us test this prediction quantitatively.

Running pairwise combinations of four interventions (RSL3/GPX4 inhibitor, SDT/exogenous ROS, FSP1 inhibitor, HDAC inhibitor/basal ROS upregulation) on 1,000 persister cells per condition and comparing against Bliss-independent predictions reveals three synergistic pairs. RSL3 + FSP1i is the strongest (Bliss synergy score 1.99): single-agent RSL3 kills 40% and FSP1i kills 3.7%, but the combination kills 84.1% — nearly double the 42.2% Bliss prediction. The pathway traces show why: with both GPX4 and FSP1 depleted, mean lipid peroxidation reaches 8.7 (near the 10.0 death threshold), whereas RSL3 alone only drives LP to 4.5 because intact FSP1 partially compensates. RSL3 + HDACi (1.25x) and FSP1i + HDACi (1.21x) are also synergistic. SDT combinations are merely additive because SDT alone already kills 100% of persisters (ceiling effect). Drug potency parameters (FSP1i 85% inhibition, HDACi 2x ROS) are estimates — exact synergy scores depend on these assumptions, but the directional finding (dual-pathway depletion > single-pathway) is robust to parameter variation.

Chapter 7

Resistance Mechanisms

7.1 Tumor Microenvironment: Oxygen Gradients and Hypoxia

The preceding simulations assumed uniform oxygenation — every cell generates the same basal ROS regardless of position. Real tumors have oxygen gradients: O2 penetrates approximately 100-150μm from the vasculature (Vaupel, Cancer Research 1989; modeled via the Krogh cylinder approximation, with underlying O2 transport physiology consistent with Anatomy & Physiology 2e, Ch.22.4), creating hypoxic cores where mitochondrial ETC activity drops and basal ROS production falls (Biology2e Ch.7, Sec.7.4: O2 is the terminal electron acceptor for the electron transport chain).

To quantify this, we applied an exponential O2 gradient from the tumor edge inward (O2(d) = exp(-d/λ), same Krogh approximation used for drug penetration) and compared treatment efficacy under uniform vs gradient conditions. Because the model only modulates basal ROS (not Fenton chemistry or SDT/PDT dose), this is a conservative estimate of hypoxia's protective effect.

The result is striking. Under the O2 gradient ($\lambda = 120\mu$m), RSL3 overall kill collapses from 3.7% to 0.1%, and in the hypoxic zone (deeper than 360μm from the tumor edge) RSL3 kills zero cells. SDT overall kill drops only modestly — from 91.9% to 87.8% — and maintains 86.6% kill even in the hypoxic zone. This differential is robust across all tested penetration lengths (λ = 80, 100, 120, 150μm) and emerges directly from the biochemistry: RSL3 inhibits GPX4 but depends on the cell's own mitochondrial ROS to initiate lipid peroxidation. In hypoxia, there is no basal ROS to drive the cascade, regardless of how thoroughly GPX4 is inhibited. SDT delivers exogenous ROS via sonosensitizer activation (primarily Type I electron-transfer mechanism, less O2-dependent than the Type II singlet-oxygen pathway), bypassing the mitochondrial bottleneck.

This finding has two implications. First, it quantifies a specific mechanism by which the tumor microenvironment protects cells from pharmacologic ferroptosis inducers —

not through upregulation of defense pathways (the NRF2/MUFA mechanisms modeled in Chapter 5), but through the absence of the upstream trigger (basal ROS). Second, it provides a mechanistic rationale for why physical ROS modalities might maintain efficacy in contexts where pharmacologic approaches fail: they supply what hypoxic cells lack. This is the same depth-vs-penetration argument made in Section 8.2, but now grounded in TME biology rather than tissue optics alone.

Caveats: the O2 model treats SDT/PDT as fully O2-independent, which is conservative (the Type II photodynamic mechanism requires O2 for singlet oxygen generation). The Fenton reaction substrate (H2O2) is also O2-derived via the superoxide→SOD pathway, but we modulate only basal_ros, not Fenton rate directly. The O2 field is steady-state (no consumption dynamics) and uses distance from the tumor edge as the O2 source (no explicit vasculature). All of these simplifications mean the actual hypoxic protection is likely STRONGER than what we report.

7.2 Immune Coupling: DAMP-Mediated T Cell Activation

Ferroptotic cell death releases damage-associated molecular patterns (DAMPs) — particularly HMGB1, calreticulin, and ATP — that activate dendritic cells and prime local T cell responses (immunogenic cell death, ICD). To explore whether the kill-density and kill-quality differential between SDT and RSL3 translates into differential immune activation, we added a spatial immune coupling layer to the TME simulation. Dead cells release DAMPs into a diffusion field; local DAMP concentration activates resident T cells via a half-maximal response function (Kd = 5.0); activated T cells kill adjacent tumor cells at rate 0.02 per step, suppressed by a PD-1 brake (70% inhibition baseline, reduced to 14% with simulated anti-PD-1).

The DAMP quality per dead cell is not equal across treatments. Lipid peroxidation is an autocatalytic free-radical chain reaction: each peroxyl radical abstracts a hydrogen from an adjacent polyunsaturated fatty acid, generating a new radical and propagating the cascade (established biochemistry; for the role of O2 as terminal electron acceptor driving basal ROS via the mitochondrial ETC, see Biology2e Ch.7, Sec.7.4). For high-ROS treatments, this cascade continues post-threshold: the simulation now continues LP accumulation for 5 steps after death (with zero repair — the cell's defenses have failed), producing emergent overshoot. SDT/PDT Persister LP reaches ∼20 (2.0× the death threshold) because exogenous ROS sustains the chain reaction, while RSL3's slower LP accumulation reaches ∼7.8. The DAMP differential therefore comes from both kill *density* (SDT kills ∼140K cells vs RSL3's ∼163) and kill *quality* (each SDT dead cell releases ∼2.6× more DAMPs).

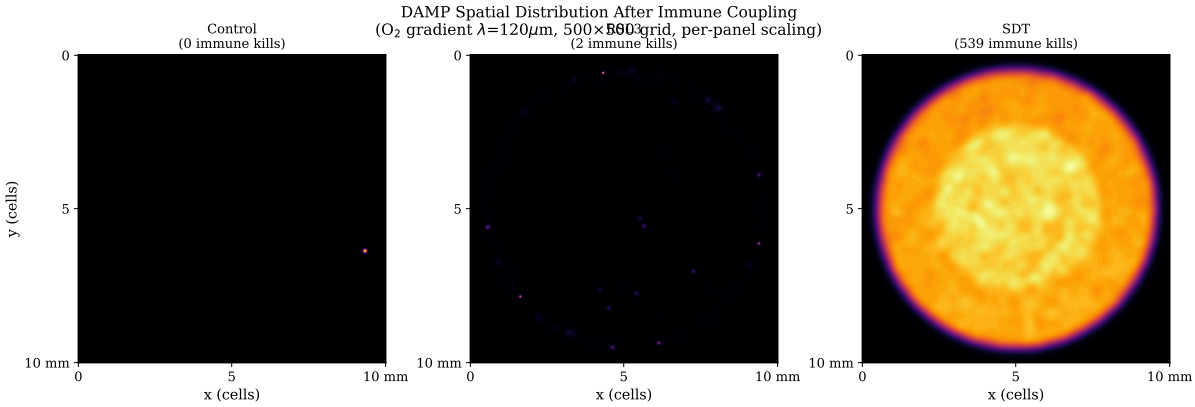

Figure 7.1: DAMP spatial distribution after immune coupling (O_2 gradient λ=120μm, per-panel scaling—intensity not comparable across panels). SDT covers the full tumor area (139,641 kills, 539 immune kills); RSL3 produces sparse isolated hotspots (163 kills, 2 immune kills).

Results (O2 gradient λ=120μm, 500×500 grid): SDT with immune coupling produces 139,640 ferroptotic kills plus 521 immune kills (88.1% total), compared to 87.8% with ferroptosis alone. Adding simulated anti-PD-1 increases immune kills to 1,477 (88.7% total) — a 2.8× amplification. RSL3 produces only 5 immune kills without anti-PD-1 and 18 with it, because its sparse kill field generates negligible local DAMP concentration. The SDT-to-RSL3 immune kill ratio is 104:1 without anti-PD-1 and 82:1 with it.

The absolute immune contribution is modest (∼0.3-0.9% additional kill for SDT), which is expected for a resident T cell model without systemic immune recruitment. The qualitative finding matters more: treatments that produce dense spatial kill patterns generate self-reinforcing immune activation that pharmacologic approaches with sparse kill cannot achieve. This suggests that the physical-modality advantage extends beyond direct ROS delivery — the spatial kill density itself creates a secondary immune mechanism that further widens the gap.

Caveats: the immune model represents resident T cell activation only (0-48h), not systemic immune recruitment or adaptive immune memory. All immune parameters (kill rate, DAMP diffusion, DC activation threshold) are estimated, not calibrated to experimental data. The LP overshoot is now emergent from the biochemistry (5 post-death steps with zero repair) rather than an estimated multiplier. PD-1/anti-PD-1 parameters are simplified (single brake coefficient). These results establish qualitative directionality — dense ferroptotic kill enables immune amplification — but the absolute magnitudes should not be treated as predictions.

7.3 Stromal Shielding: CAF-Mediated Protection at the Tumor Periphery

The preceding TME features modeled the tumor interior (hypoxia, immune coupling). But the tumor boundary presents a different biology: cancer-associated fibroblasts (CAFs) at the tumor-stroma interface supply neighboring tumor cells with cysteine (for GSH synthesis via the GGT1/SLC7A11 pathway) and oleic acid (MUFA for membrane remodeling via ACSL3-mediated uptake). This metabolic cooperation has been documented in CAF co-culture studies **?** and lipid-ferroptosis research **?**; **?**, but has not been quantified in a spatial ferroptosis simulation.

To test whether stromal shielding meaningfully affects treatment outcomes, we identified the 1,796 tumor cells (1.1% of 159,033 total) with at least one stromal Moore neighbor and applied per-step CAF-mediated boosts: GSH +0.06/step (2.4× the endogenous NRF2 recovery rate, capped at 18.0 mM) and MUFA +0.003/step (capped at 0.25). These are estimated parameters — no textbook coverage exists for CAF biology in any of the reference textbooks used in this project. The underlying biochemistry (GSH scavenging, MUFA-mediated PUFA displacement) is handled by the existing ferroptosis-core engine; the CAF contribution simply increases the supply of two defensive metabolites to boundary cells.

The result is treatment-selective. Among the 1,796 stromal-adjacent cells, CAF protection halves the RSL3 kill rate from 3.0% to 1.5% (50% relative reduction) but reduces the SDT kill rate from only 96.1% to 91.2% (5.1% relative). The hypoxic zone kill rate is unchanged for all treatments — CAFs are at the boundary, not the interior. The overall kill rate change is small (∼0.06% for SDT, ∼0.02% for RSL3) because only 1.1% of tumor cells are affected.

The mechanism behind the treatment selectivity is the same ROS imbalance that drives the hypoxia finding: RSL3 depends on slow basal ROS (∼0.3 relative units) to drive the LP cascade through compromised GPX4. A 2.4× GSH supply boost meaningfully slows this cascade — each step, more ROS is scavenged before it reaches lipid membranes. SDT delivers ∼17× higher exogenous ROS (∼5.0 relative units), overwhelming the CAF defense within a few steps regardless of the GSH boost. The Michaelis-Menten scavenging kinetics (GSH of 18 provides only 26% better scavenging than GSH of 5) ensure that even maximal CAF supply cannot rescue cells from SDT's ROS flood.

This finding complements the hypoxia and immune coupling results: physical modalities bypass both hypoxic protection (Section 7.1) and CAF-mediated stromal shielding, while also generating stronger downstream immune activation (Section 7.2). Hypoxia and stromal shielding are direct resistance mechanisms that blunt pharmacologic ferroptosis induction; immune coupling is a separate amplification effect that favors treatments with dense spatial kill patterns.

Caveats: all stromal parameters are estimated, not calibrated to experimental data. The model treats all boundary stromal cells as equally protective CAFs (in reality, CAF density and phenotype vary). CAF metabolic exhaustion is not modeled — the supply rate is constant regardless of demand. ECM-mediated drug diffusion barriers are not included (deferred to a future iteration). Only the outermost ring of tumor cells (depth = 1 cell layer) receives CAF protection; paracrine signaling range may extend further. These results establish that stromal shielding is treatment-selective in the model, but the absolute magnitudes depend on estimated parameters.

7.4 Acidic pH: Iron Release vs Drug Ion Trapping

Tumor cores are acidic (pH 6.5-6.8) due to lactic acid accumulation from aerobic glycolysis (the Warburg effect — not covered in the reference textbooks, which associate lactic acid fermentation with anaerobic conditions only). This acidity creates two competing effects on ferroptosis: (1) ferritin protein shells destabilize below pH ∼6.5, releasing stored iron into the labile iron pool and increasing Fenton-driven ROS generation (pro-ferroptosis); (2) weak-base drugs become protonated at low extracellular pH and are trapped in acidic compartments, reducing intracellular drug availability (anti-ferroptosis, governed by Henderson-Hasselbalch equilibria — Chemistry2e Sec.14.2). The net effect on ferroptosis is not obvious from first principles: does the iron boost or the drug loss dominate?

To quantify this, we applied an exponential pH gradient from the tumor edge inward ($pH(d) = 7.4 - 0.9 \times (1 - \exp(-d/\lambda))$, λ=120μm), yielding pH 7.4 at the well-perfused periphery and ∼6.5 in the deep core. The iron modulation (2.35× at pH 6.5) increases the Fenton ROS term, while the ion trapping (36% drug loss at pH 6.5, linearized Henderson-Hasselbalch) reduces RSL3's effective GPX4 inhibition from 92% to 59%.

The result is unambiguous: drug trapping dominates. RSL3 ferroptosis kills drop from 163 to 77 (53% reduction) because the GPX4 restoration (from 8% to 41% of normal activity) is a 5.1× repair boost, while the iron-driven ROS increase is only 7%. SDT is barely affected (+0.8%, from 139,640 to 140,693 ferroptosis kills) because ultrasound energy delivery is pH-independent — there is no drug to trap — and the iron boost adds only 0.4% to SDT's total ROS (dwarfed by exogenous ROS of 5.0 relative units). Control shows a tiny increase (1 to 2 ferroptosis kills) from the Fenton boost alone.

This adds a third direct resistance mechanism — alongside hypoxia (Section 7.1) and stromal shielding (Section 7.3) — that selectively penalizes pharmacologic ferroptosis inducers while leaving physical modalities unaffected. All three share the same structural asymmetry: pharmacologic approaches depend on endogenous processes that the TME can disrupt — basal ROS generation (removed by hypoxia), GSH depletion (counteracted by stromal cysteine supply), and intracellular drug bioavailability (reduced by pH-driven

ion trapping) — while physical modalities deliver exogenous energy that bypasses these vulnerabilities.

Caveats: all pH parameters are estimated. RSL3 is a chloroacetamide, not a classic weak base — its pKa and ion trapping behavior are poorly characterized, making the ion_trap_sensitivity the most uncertain parameter in this model. The linear Henderson-Hasselbalch approximation is valid over the narrow pH 6.5-7.4 range but would break down at more extreme pH values. The pH field is steady-state (no lactate production or buffering dynamics). Tumor pH ranges (6.5-6.8) are from primary literature; no reference textbook covers tumor-specific pH data.

Chapter 8

Counterarguments and Failure Modes

8.1 Other Directions The Repo Was Underweighting

One reason to widen the article framing is that the repo is no longer only telling a PDT/SDT story. Even under conservative tagging, several adjacent directions deserve explicit mention because they may compete with or complement the physical-ROS hypothesis.

Radioligand therapy is the clearest example of a clinically advancing, spatially selective modality-class competitor. It shares the intuitive appeal of localized tumor killing while operating within a much more mature theranostic and dosimetry framework.

Immune residual-disease strategies — especially TIL/TCR-style cell therapy, therapeutic mRNA or neoantigen vaccination, and phagocytosis-checkpoint manipulation — are important because they aim not just to kill a vulnerable state, but to convert residual disease into an immune-recognizable target. The revised mRNA-vaccine tagging pass is a concrete reminder that this lane must be interpreted through therapeutic-cancer-vaccine papers rather than the broader post-COVID vaccine literature.

Pathway-target layers such as SCD/MUFA remodeling, DHODH-linked backup defenses, DHCR7/7-DHC biology, and TRIM25/GPX4 degradation matter because they may explain why ferroptosis-looking vulnerabilities repeatedly weaken in vivo.

Stromal shielding, senescence programs, and broader tumor-microenvironment biology are not side notes. They may be the main reason apparently elegant modality-level ideas stall during translation. The stromal protection simulation (Section 7.3) now quantifies one component of this: CAF-mediated metabolic support halves RSL3's peripheral kill rate while leaving SDT largely unaffected.

The practical implication is that resistant-state matching should become the primary decision layer. Modality choice matters, but it should come after the biology of residual survival is defined.

8.2 The Modality Question: PDT, SDT, and the Tissue-Penetration Tradeoff

If OXPHOS-resistant persister cells are ferroptosis-vulnerable, which modality best exploits this? Our initial corpus analysis identified SDT as unusually aligned with ferroptosis among physical ablative modalities. However, broader PubMed verification reveals a critical context we must address honestly:

Table 8.1: Ferroptosis engagement across physical modalities (PubMed, March 2026).

Modality	Ferroptosis	Ferro+ICD	Depth
PDT	**355**	**67**	Superficial (mm)
SDT	**121**	**25**	Deep (cm)
IRE	15	emerging	Invasive
HIFU	3	minimal	Deep (cm)
TTFields	0	0	Surface

PDT dominates SDT on every ferroptosis and ICD metric by approximately 3:1. This fundamentally challenges any claim that SDT is uniquely positioned for ferroptosis-ICD. PDT — using light rather than ultrasound to activate sensitizers — triggers the same ROS → GSH depletion → ferroptosis → ICD chain with a larger evidence base, more clinical experience, and approved photosensitizers.

SDT's remaining advantage is tissue penetration. Ultrasound penetrates centimeters into tissue; light penetrates millimeters. For deep-seated tumors (pancreatic, hepatic, pelvic), SDT can reach targets that PDT cannot. For accessible superficial tumors (skin, oral, esophageal, bladder surface), PDT is the established modality and SDT offers no advantage.

The honest framing is therefore: SDT is not the unique ferroptosis-ICD modality — PDT is the established one. SDT is a **depth-extended variant** of the same photosensitizer-based ROS-ferroptosis-ICD approach, using ultrasound instead of light to reach tumors that PDT cannot access. This is a meaningful clinical advantage for deep tumors, but it is incremental, not paradigmatic.

The mechanism is shared: sensitizer-activated ROS → GSH depletion → GPX4 inactivation → ferroptosis → DAMP release → ICD Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; J et al. [2021]. SDT-specific data shows 72 GSH/GPX4 articles (11.7% of corpus) and 6 dual-pathway articles J et al. [2021]; R et al. [2024]; H et al. [2025b]. A recent Nature Communications paper demonstrates that IRE can also be combined with ferroptosis via iron-based nanoparticles L et al. [2026], further indicating that the ferroptosis-physical connection is broader than SDT alone.

Critical caveats remain: Most SDT-ferroptosis data involves engineered nanosonosensitizers Q et al. [2021]; Y et al. [2021b]. No clinical study has verified SDT-induced ferrop-

tosis or ICD in human tumors. The PDT precedent — 40 years of development without demonstrating robust ICD-mediated immune synergy in randomized trials — applies as a direct cautionary parallel.

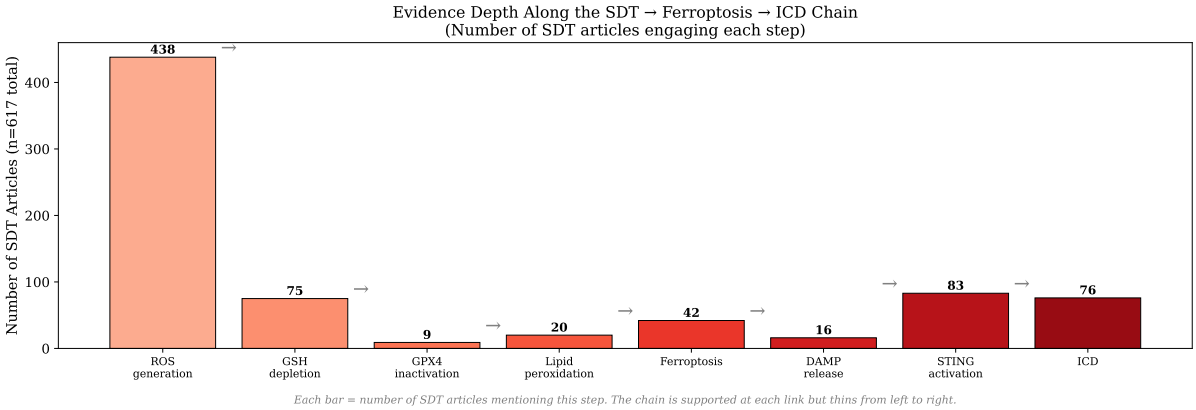

Figure 8.1: SDT ferroptosis-ICD chain evidence.

8.3 Counterarguments, Precedents, and Reasons This Could Fail

The photodynamic therapy precedent. PDT is conceptually identical to SDT: physical energy (light vs ultrasound) activates a sensitizer to generate ROS, causing cell death with ICD potential. PDT has been in clinical development for 40+ years and has achieved approval for a few indications (esophageal, skin) but has never demonstrated the "ROS → ICD → immune synergy → durable response" chain in a randomized clinical trial. PDT + immunotherapy combinations remain at the case report level Y et al. [2026]. If this logic worked robustly, PDT would have shown it by now. SDT's claimed advantages over PDT — deeper tissue penetration and potentially superior ferroptosis engagement via engineered nanosonosensitizers — are plausible but incremental, not paradigmatic. The SDT hypothesis must explain why it would succeed where PDT has not demonstrated clear immune synergy.

The ferroptosis vulnerability window may be transient. Persister cells downregulate FSP1 as part of the mesenchymal/dedifferentiated state M et al. [2026]. However, persisters are a transitional population: they eventually either die or re-enter proliferation with restored drug resistance. The ferroptosis-sensitive window may last days to weeks. Our current simulation suggests that exogenous ROS might keep a longer effective window open than pharmacologic GPX4 inhibition alone, but that result is still model-dependent and does not yet incorporate the full in vivo lipid-remodeling problem. If the real window closes before SDT or PDT can be administered, the therapeutic opportunity is lost.

The lipid-remodeling and backup-defense problem. OXPHOS-resistant cells may co-upregulate NRF2-driven antioxidant defenses, SCD/MUFA remodeling, DHODH-linked backup defenses, DHCR7/7-DHC biology, TRIM25-linked GPX4 degradation dynamics, and other lipid-handling programs that neutralize ferroptosis vulnerability even as the metabolic preconditions accumulate. The pathway-target layer now shows that these escape axes are present in the corpus, but mostly in review-heavy or early mechanistic form. They therefore strengthen the need for caution more than they strengthen any single therapeutic claim.

The nanosonosensitizer confound. Most SDT-ferroptosis papers use engineered nanoparticles *designed* to deplete GSH and release iron. If plain SDT with conventional sonosensitizers doesn't trigger ferroptosis, the insight is about nanoparticle design, not SDT as a modality — and the regulatory path becomes device+drug+nanoparticle, the most complex category.

The ICD-to-immunity chain has repeatedly underperformed clinically. PexaVec (oncolytic virus ICD, failed Phase 3) M et al. [2019], CheckMate-498 (immunotherapy in GBM, failed) A et al. [2023], KEYNOTE-361 (ICI+chemo in urothelial, failed) T et al. [2021], and BIND-014 (nanoparticle, failed Phase 2) Von Hoff et al. [2018] all demonstrate that preclinical rationale frequently fails in translation. SDT has no reason to be exempt from this base rate.

Narrow applicability. The physical-ROS hypothesis applies only to accessible or image-localizable solid tumors, immunocompetent settings where ICD could matter, tumors that actually enter a ferroptosis-prone residual state, and a specific post-therapy time window. Hematologic malignancies, diffuse metastatic disease, and many non-localizable residual states fall outside its plausible range.

Evolutionary limitation. This hypothesis addresses one resistance transition. After SDT kills ferroptosis-sensitive persisters, survivors are selected for ferroptosis resistance (GPX4 upregulation, NRF2 activation, FSP1 restoration). The approach adds one step to the therapeutic arms race but does not end it. Durable eradication would require the ICD-triggered immune response to eliminate residual heterogeneous disease — the standard hope of all immunotherapy combinations, not specific to this hypothesis.

Prior art. Collateral sensitivity (Pluchino et al., 2012), adaptive therapy (Gatenby & Brown, 2009), and synthetic lethality are established frameworks for exploiting resistance tradeoffs. What our analysis adds is the specific SDT connection, not the general principle.

Part IV

What Should Happen Next

Chapter 9

Research Roadmap

9.1 Research Priorities

Priority 1 (highest value, ∼3 months): Comparative resistant-state killing experiment. Using persister or residual-disease models grounded in the current literature M et al. [2026], compare PDT, SDT, and pharmacologic ferroptosis inducers under matched conditions. Measure cell death, lipid-peroxidation markers, and ICD outputs (calreticulin, HMGB1, STING). Include conventional sensitizers and nanosensitizer conditions so the modality-versus-nanoplatform question is not blurred.

Priority 2 (∼1 month): Measure the vulnerability window and backup defenses. After drug withdrawal from persister-inducing therapy, quantify GPX4, FSP1, NRF2, SCD/MUFA markers, and ferroptosis sensitivity at serial timepoints. If the window closes rapidly or lipid remodeling rises immediately, the translational case weakens substantially.

Priority 3 (∼2-3 months): Compare the physical-ROS hypothesis against other adjacent lanes rather than against nothing. Radioligand therapy, checkpoint-plus-myeloid strategies, and pathway-target approaches should be evaluated on the same axes: tissue reach, evidence maturity, delivery burden, and biological plausibility for residual disease.

Priority 4 (if Priorities 1-3 are encouraging, ∼6 months): Move to in vivo comparison with honest comparators. Any mouse study should include PDT + anti-PD-1, SDT + anti-PD-1, and a pharmacologic ferroptosis arm; otherwise the experiment cannot tell whether SDT offers a meaningful advantage over its closest precedent.

Chapter 10

The Broader Landscape

10.1 Conclusion

This article began as a modality-centered question and ends in a broader place.

What the corpus analysis shows. Cancer-therapy literatures need to be compared on resistant-state relevance, evidence maturity, delivery constraints, and known analytic artifacts before any one mechanism is over-privileged. Our evidence tagger has 96% precision but only 55% recall — meaning tagged evidence is almost always real, but absence claims require qualification. The tissue-of-origin layer reveals that 74% of physical-modality research is concentrated in epithelial tumors, with hematologic and mesothelial contexts nearly absent. The designed-combination classifier shows that only 19.7% of multi-mechanism papers represent actual combination studies — the rest are reviews or incidental co-mentions.

What the simulations show. The mechanistic simulations produce three findings that, if validated experimentally, would have translational implications:

1. *Drug combination synergy through dual-pathway depletion.* RSL3 (GPX4 inhibitor) + FSP1 inhibitor produces 1.99× Bliss synergy because depleting both parallel repair pathways drops total antioxidant defense below the autocatalytic LP propagation threshold. This suggests combination strategies targeting GPX4 AND FSP1 simultaneously, rather than either alone.

2. *The tumor microenvironment selectively favors physical over pharmacologic ferroptosis inducers — through three resistance mechanisms and one amplification effect.* (a) Hypoxia: RSL3 kill collapses from 3.7% to 0.1% under oxygen gradients; SDT maintains 87.8%. (b) Stromal shielding: CAF-mediated GSH/MUFA supply halves RSL3's peripheral kill rate (3.0% to 1.5%); SDT barely affected (96.1% to 91.2%). (c) Acidic pH: ion trapping halves RSL3 ferroptosis kills (163 to 77, 53% reduction) as drug bioavailability drops 36% at pH 6.5; SDT gains only 0.8% (iron boost negligible vs exogenous ROS). These three resistance mechanisms share the same structural asymmetry: pharmacologic

inducers depend on slow endogenous processes that the TME disrupts, while physical modalities deliver exogenous energy that bypasses these vulnerabilities. (d) Immune coupling is a downstream amplification that also favors physical modalities: SDT generates 104× more immune kills than RSL3 (521 vs 5), amplified 2.8× by anti-PD-1 (to 1,477).

3. *Tissue-specific drug penetration creates a substantial in-vitro-to-in-vivo translation gap.* Even at the blood vessel wall, tissue-specific vascular permeability reduces RSL3-like drug kill from 40% (2D culture) to 12.1% (well-vascularized epithelial) to 2.6% (poorly-vascularized pancreatic) to 1.8% (CNS/BBB). The gap is even larger once the penetration gradient into the tissue is considered.

What remains conditional. Within the broader frame, the hypothesis that physical ROS-generating modalities may be useful for locally targetable, ferroptosis-prone residual states survives not because we've proven it, but because the simulation identifies specific conditions under which it holds (hypoxic cores, deep tissue, limited drug penetration) and specific conditions under which it doesn't (hematologic malignancies, tumors without ferroptosis-prone states, settings where ICD-to-immunity chain has historically underperformed). PDT is the established precedent, SDT is the depth-extended variant, and all are constrained by nanosensitizer dependence, uncertain in vivo ferroptosis durability, and the historical failure rate of preclinical immune logic in trials.

The deeper lesson is methodological. Gap claims can be artifacts. Mechanism counts are not maturity. Patient-study signal is not phase evidence. A compelling literature bridge is still only the start of a research program. But the simulation work adds something that literature analysis alone cannot: it identifies which biological conditions would need to be true for the hypothesis to work, and which would falsify it. The RSL3+FSP1i synergy, the hypoxia differential, and the tissue penetration gap are all testable in the lab.

Openness and reproducibility. All code, data, and analysis pipelines are openly available under the MIT license. The ferroptosis-core simulation engine is published as a standalone Rust library with Python bindings, so any researcher can reproduce, challenge, or extend the computational work. We believe this kind of radical openness — where the tools are as available as the conclusions — is essential for research that aims to help people rather than to be right about one hypothesis.

The right next step is therefore not to defend one favored modality, but to run the comparative experiments and cross-lane evaluations that could falsify it quickly.

Chapter 11

How To Contribute

[To be added in #103]

Part V

References and Tools

[Appendices A-C to be added in #104]

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